TREE OF SOULS

Final Project Information

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CS 396

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GitHub: https://github.com/Emily-McNett/ShaderFinalProject/tree/main

Brief Description

On the “FinalLandscape” level of my final project I have created an Avatar – Tree of Souls inspired scene. The scene features a procedurally generated tree created with Blender Geometry Nodes. The tree is covered in emissive vines made with a Niagara System. The large ferns placed throughout the scene are added with a PCG Graph, while the smaller plants were added with a foliage painter. The materials for the plants were created by altering imported textures. The blue paths were painted onto the landscape with the landscape painter after the texture was created in and exported from substance designer. The fog was created with a volumetric material and the darker sky/atmosphere and stars were added with a BP_Sky_Sphere.

Final Visual References

Images

Idea

The Tree of Souls from Avatar the way I remember it. This being a separated area with a large tree covered in emissive vines and a path system through the vines surrounded by a range of jungle plants.

Tutorials and Sources Used

Blender Tutorials

Blender - Procedural Trees with Geometry Nodes - Part 1/3

Blender - Procedural Trees with Geometry Nodes - Part 2/3

Plant Models

Fern | Fab

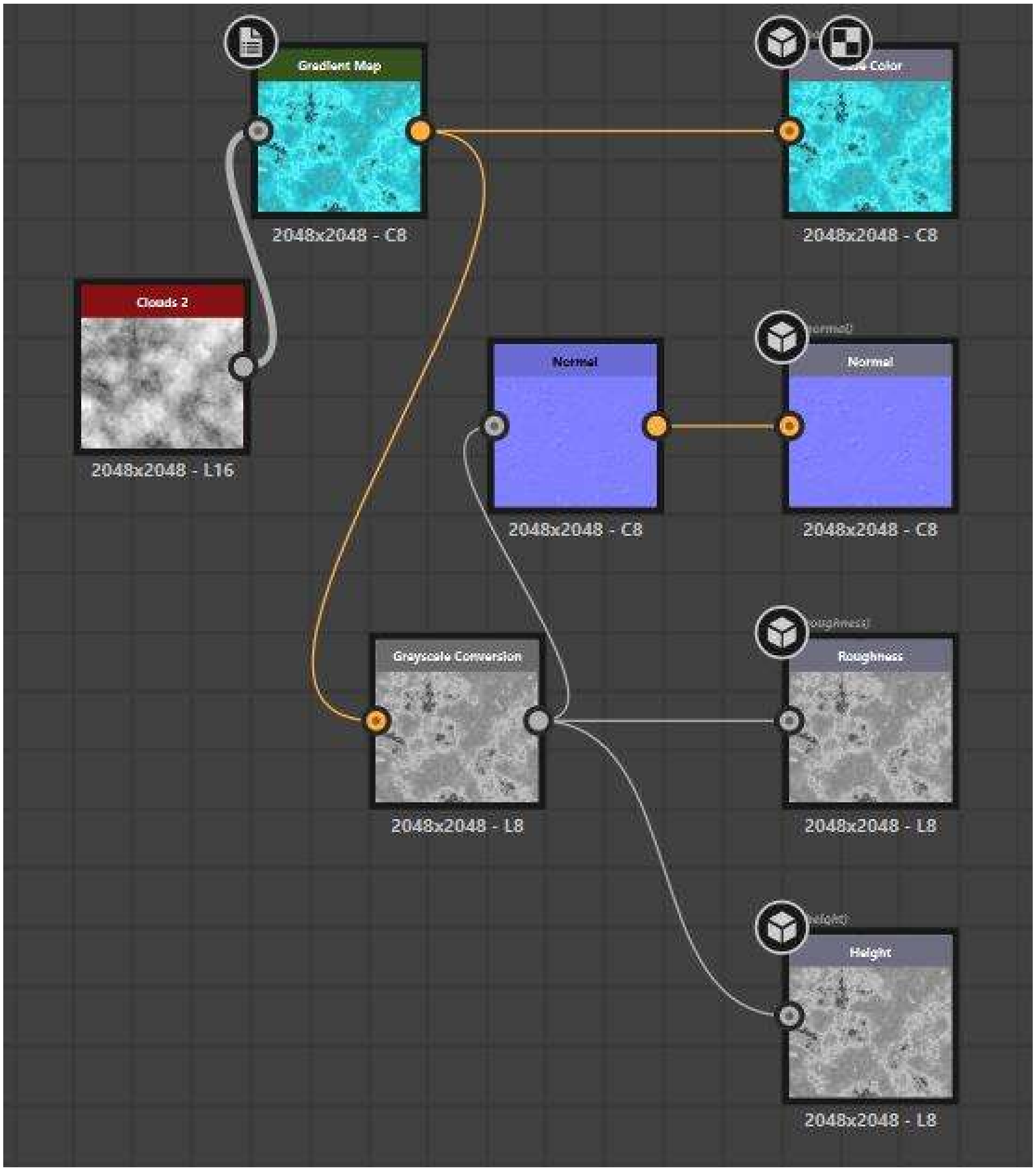
Fern | Fab - Another Fern

Beech Fern | Fab

Grass Clumps | Fab

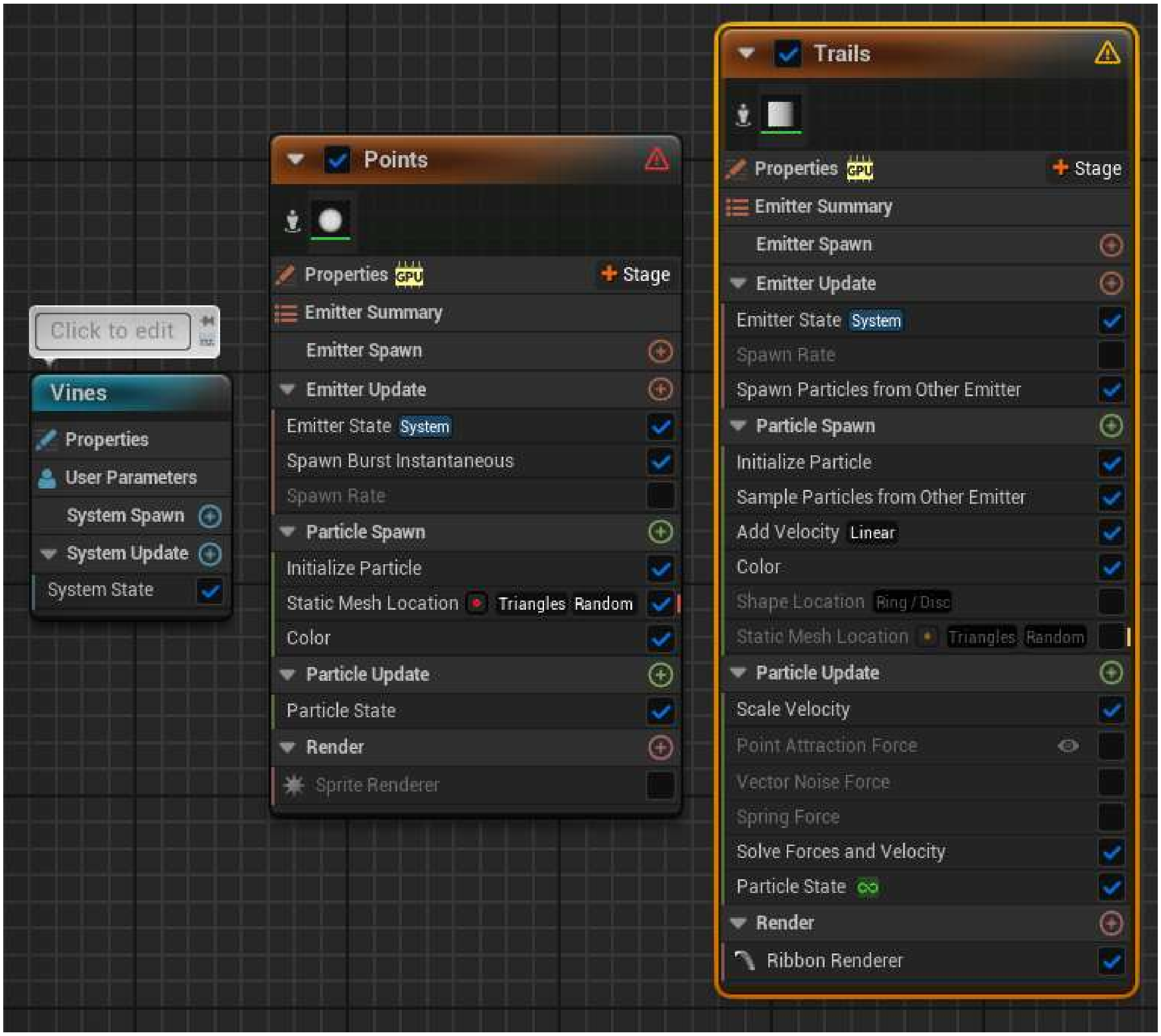
Tree Bark | Fab

Substance Designer – Path

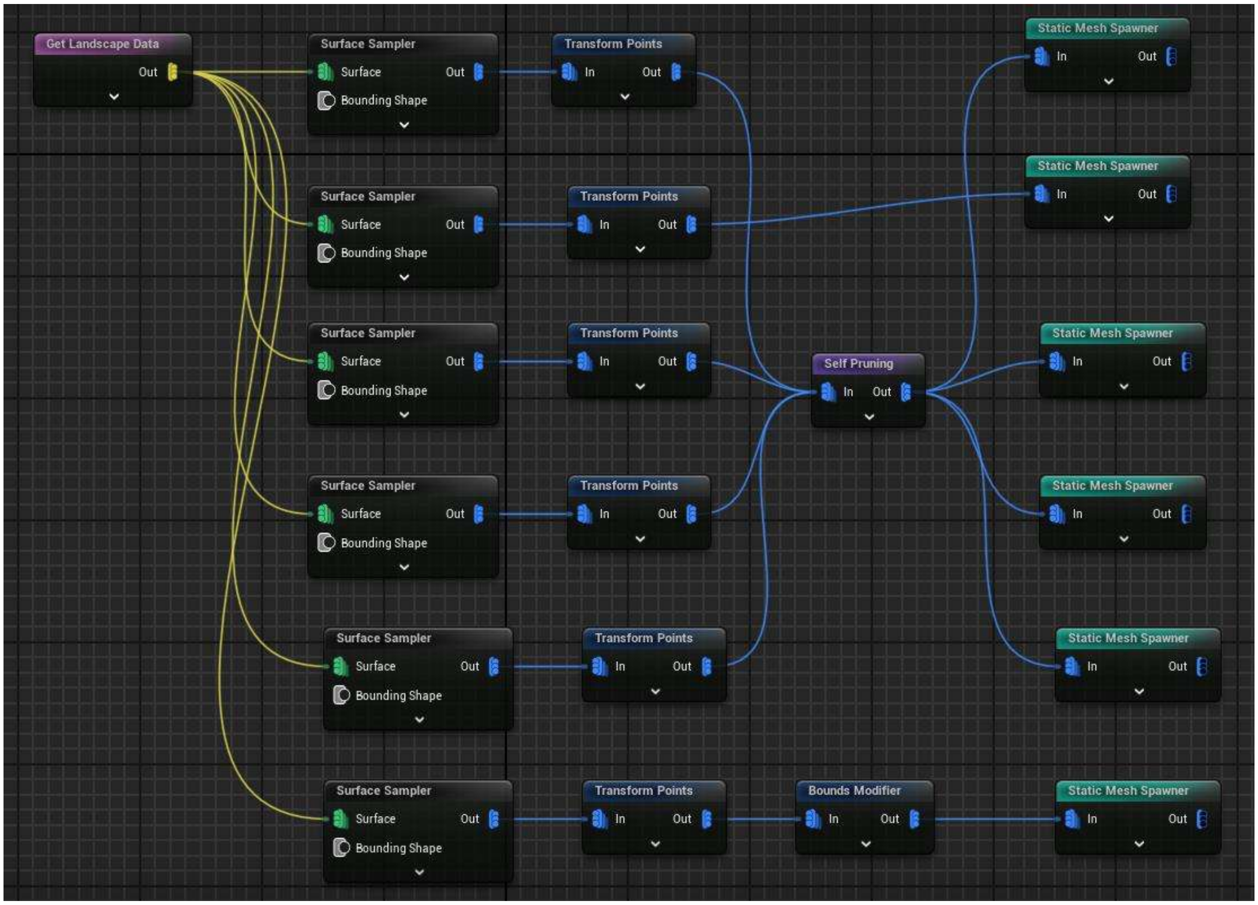


UE5 Graphs

Niagara System - Vines

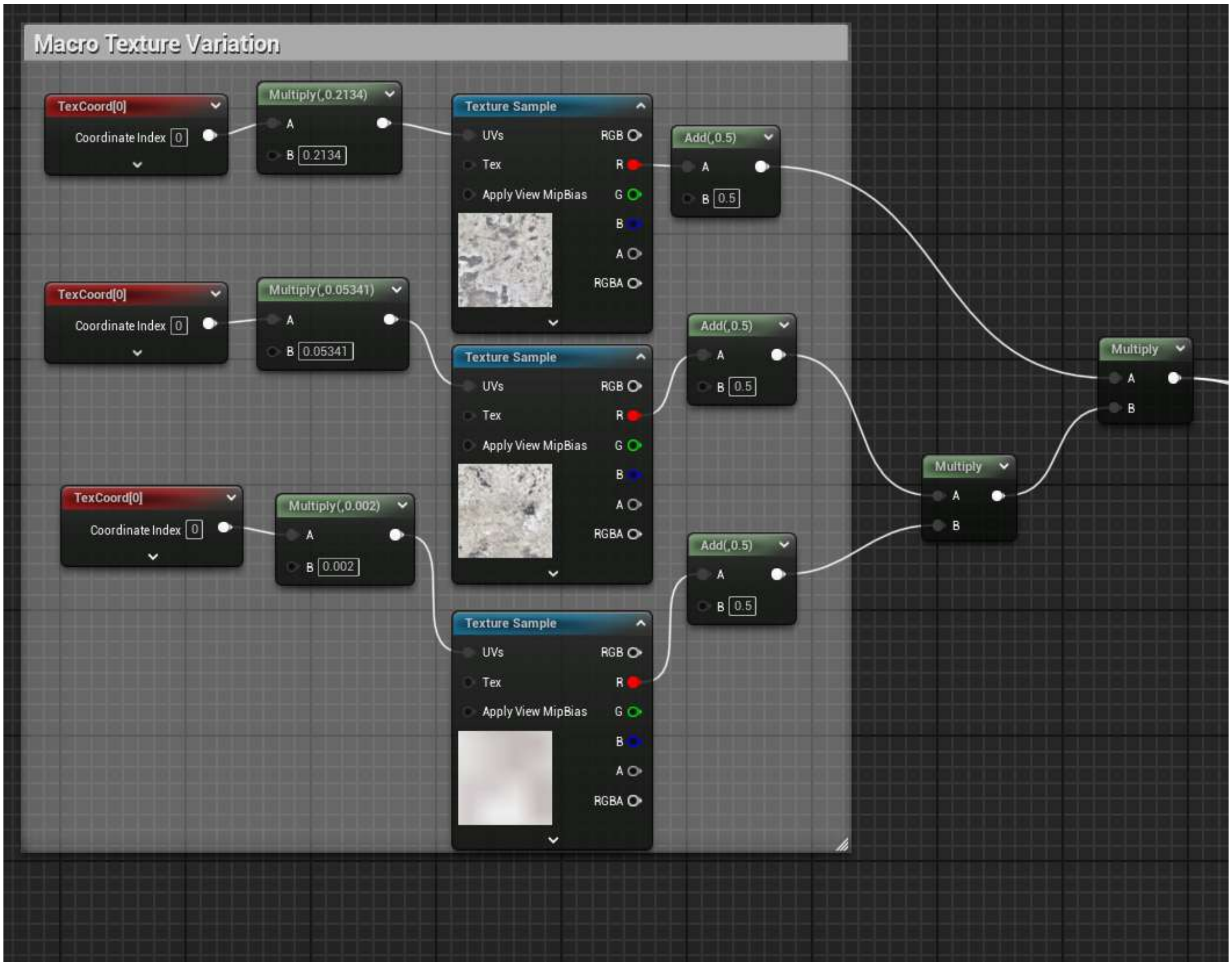


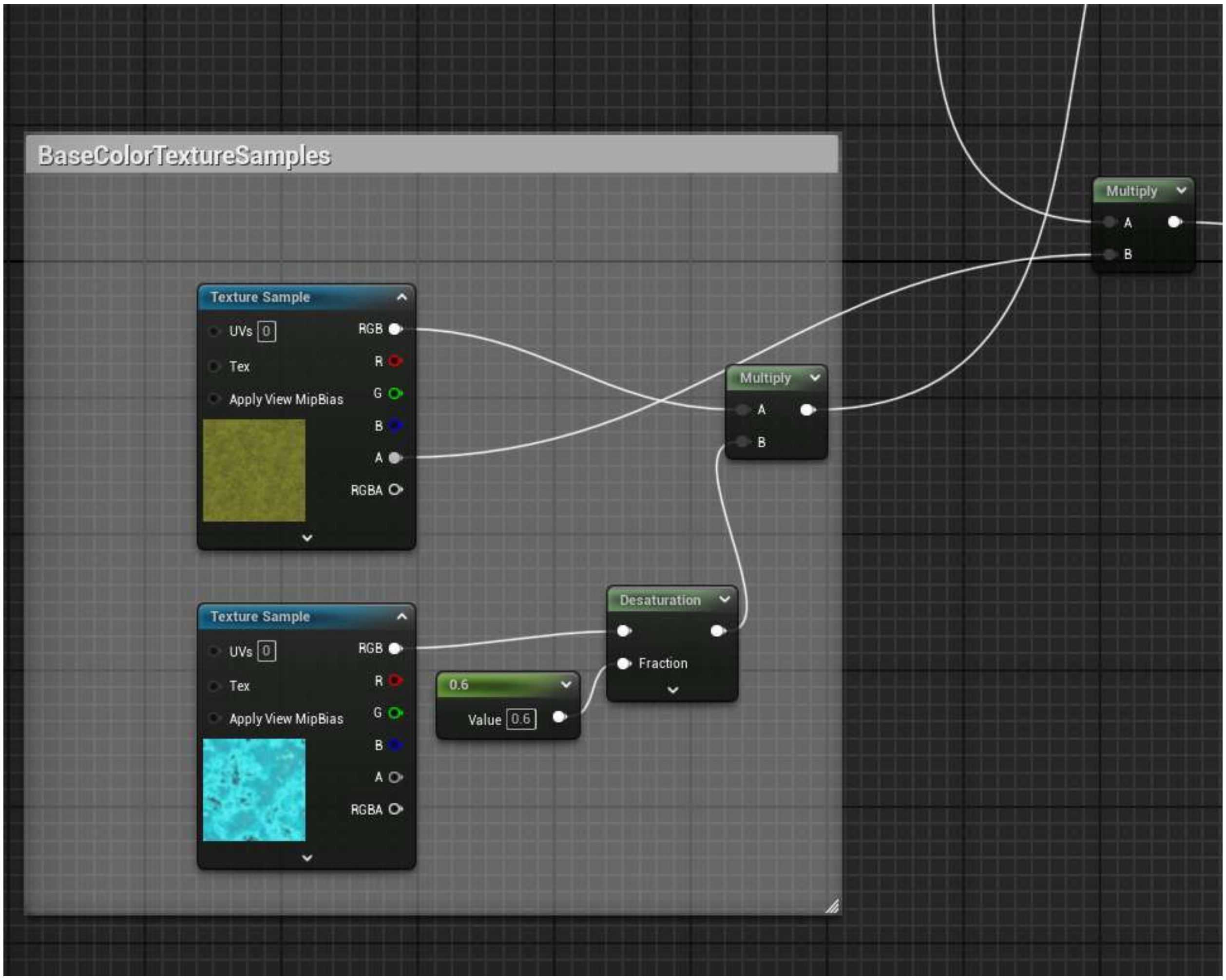
PCG graph – Large Ferns

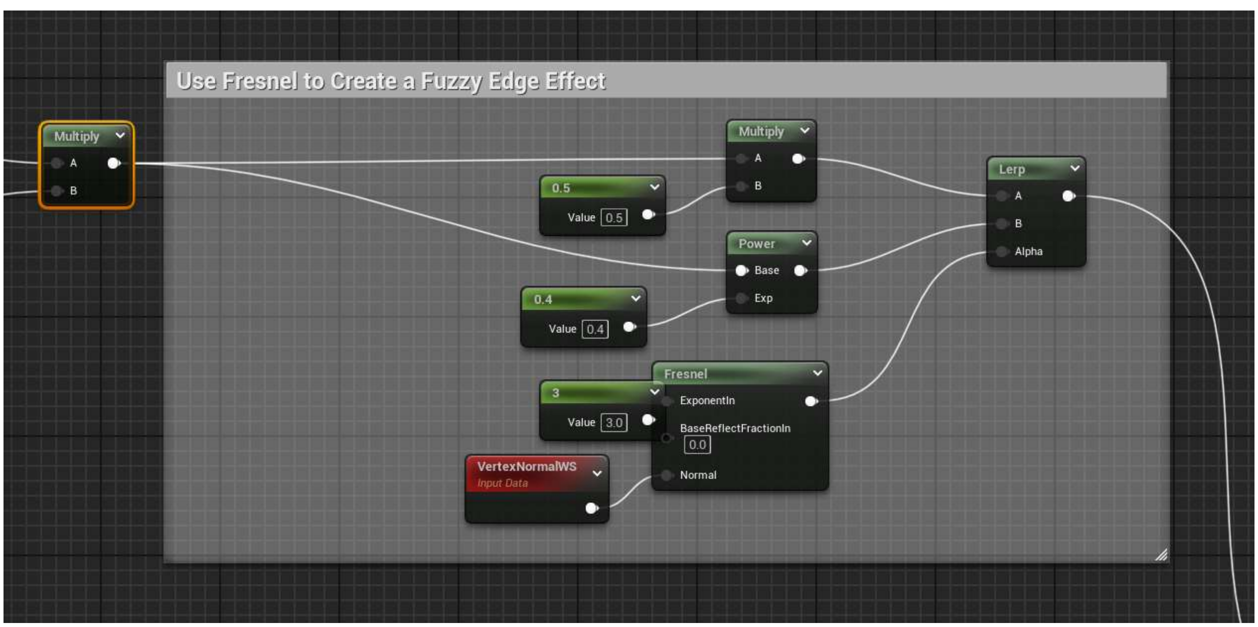


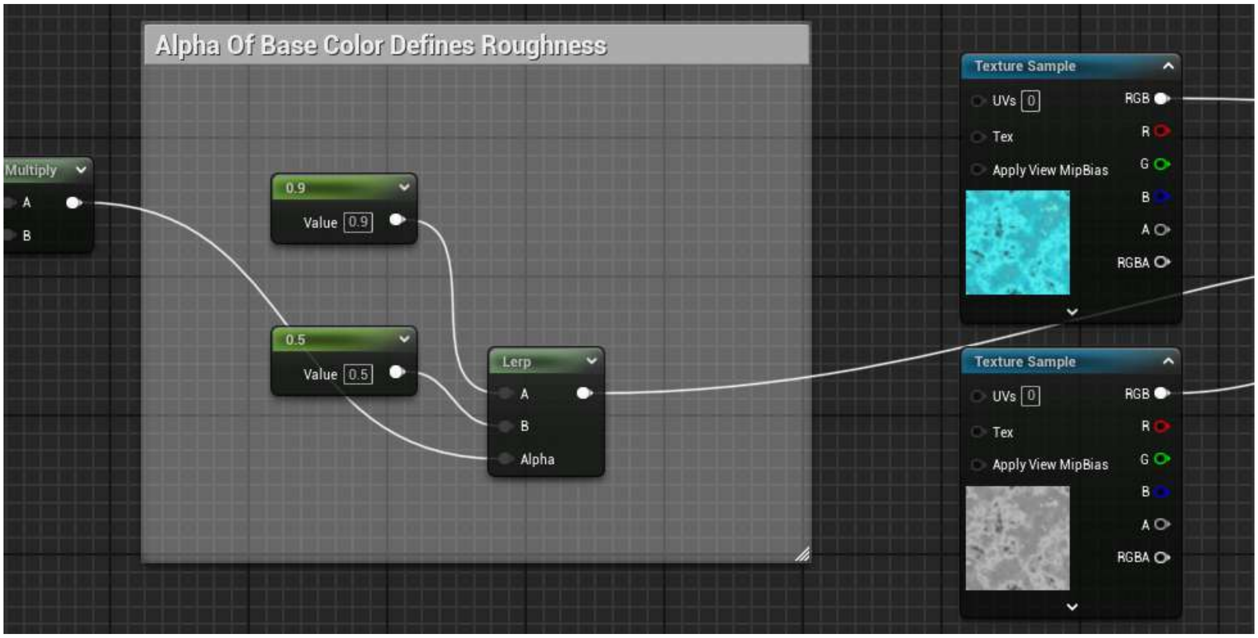
Landscape

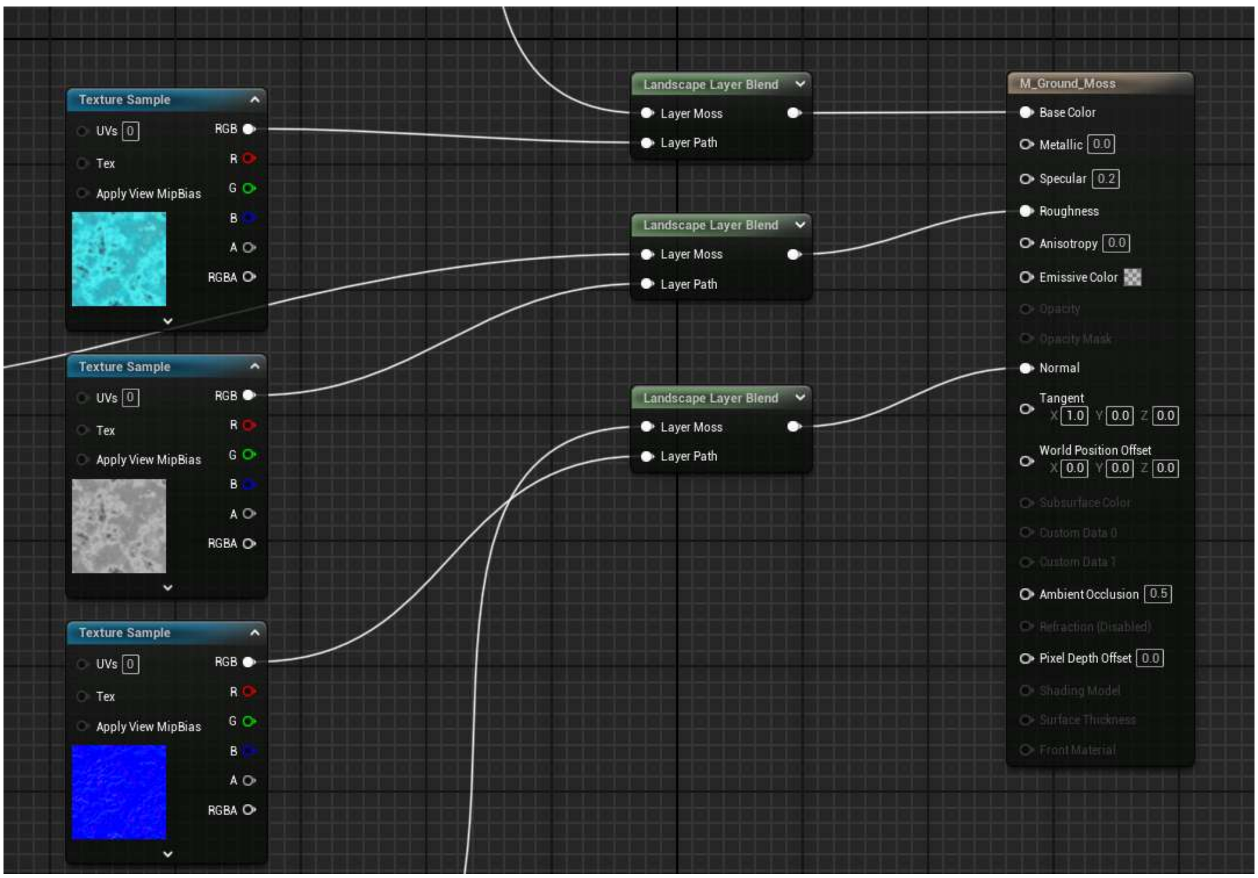
- Altered a default texture in order to add painted paths

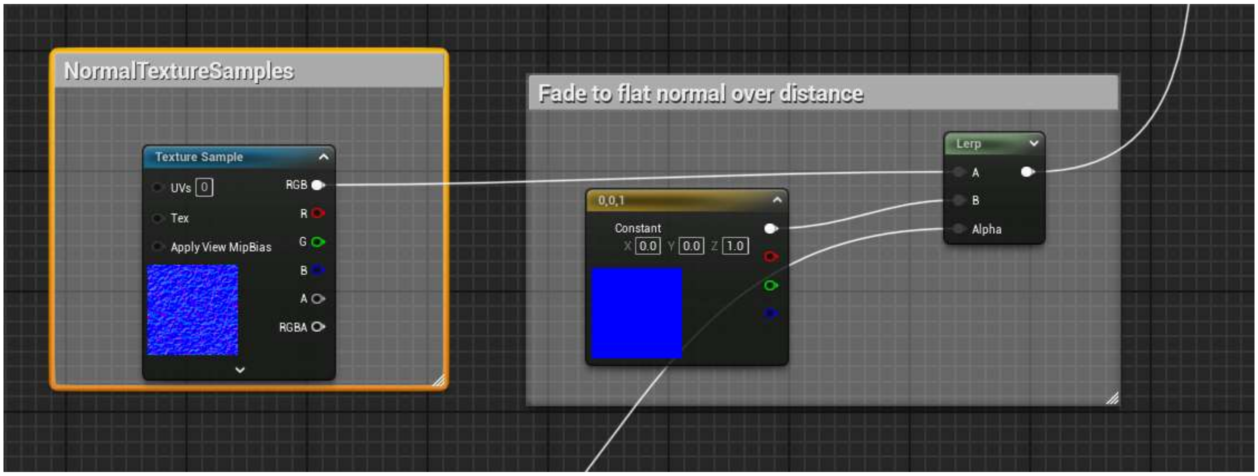

Macro Texture Variation
TexCoord[0]
Coordinate Index 0
Multiply(,0.2134)
A
B 0.2134
Texture Sample
UVs
Tex
Apply View MipBias
RGB
R
G
B
A
RGBA
Add(,0.5)
A
B 0.5
TexCoord[0]
Coordinate Index 0
Multiply(,0.05341)
A
B 0.05341
Texture Sample
UVs
Tex
Apply View MipBias
RGB
R
G
B
A
RGBA
Add(,0.5)
A
B 0.5
TexCoord[0]
Coordinate Index 0
Multiply(,0.002)
A
B 0.002
Texture Sample
UVs
Tex
Apply View MipBias
RGB
R
G
B
A
RGBA
Add(,0.5)
A
B 0.5
Multiply
A
B
Multiply
A
B

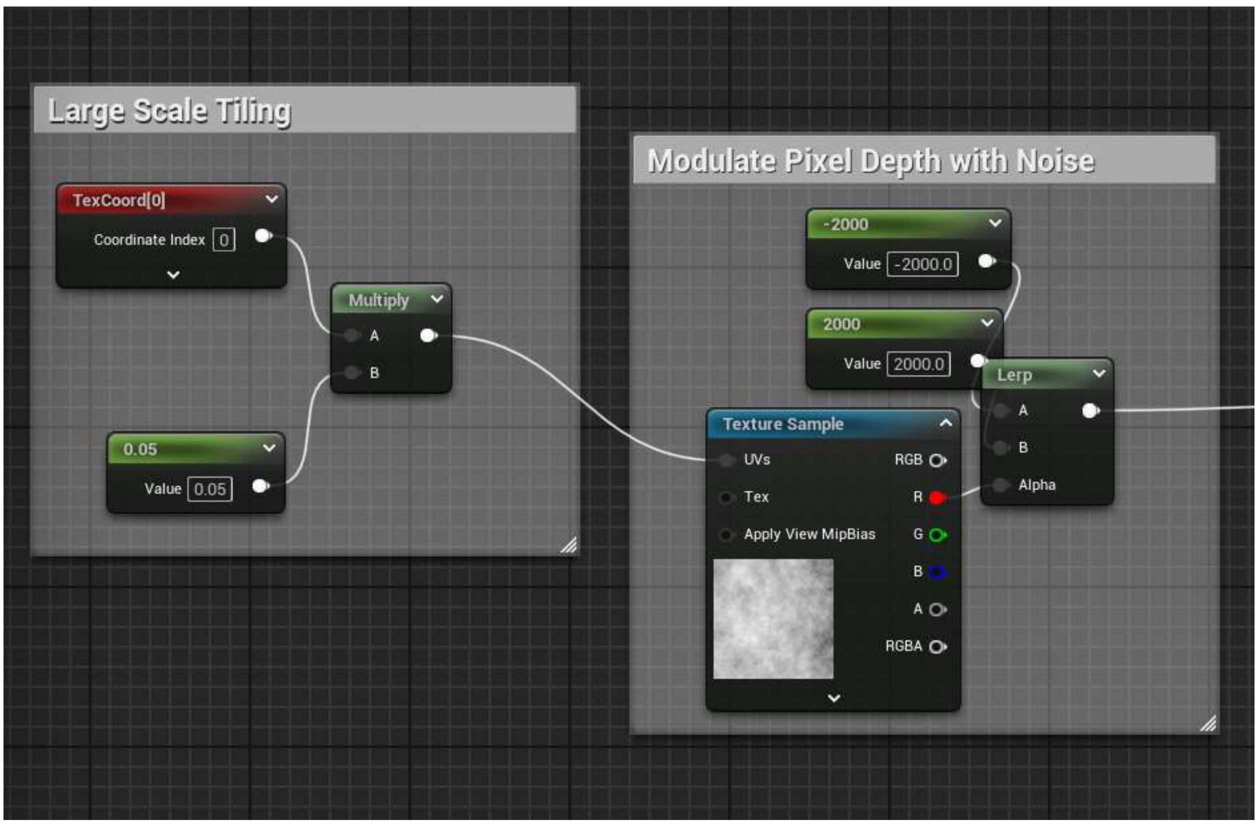

BaseColorTextureSamples
Texture Sample
UVs 0
Tex
Apply View MipBias
RGB
R
G
B
A
RGBA
Texture Sample
UVs 0
Tex
Apply View MipBias
RGB
R
G
B
A
RGBA
0.6
Value 0.6
Desaturation
Fraction
Multiply
A
B
Multiply
A
B

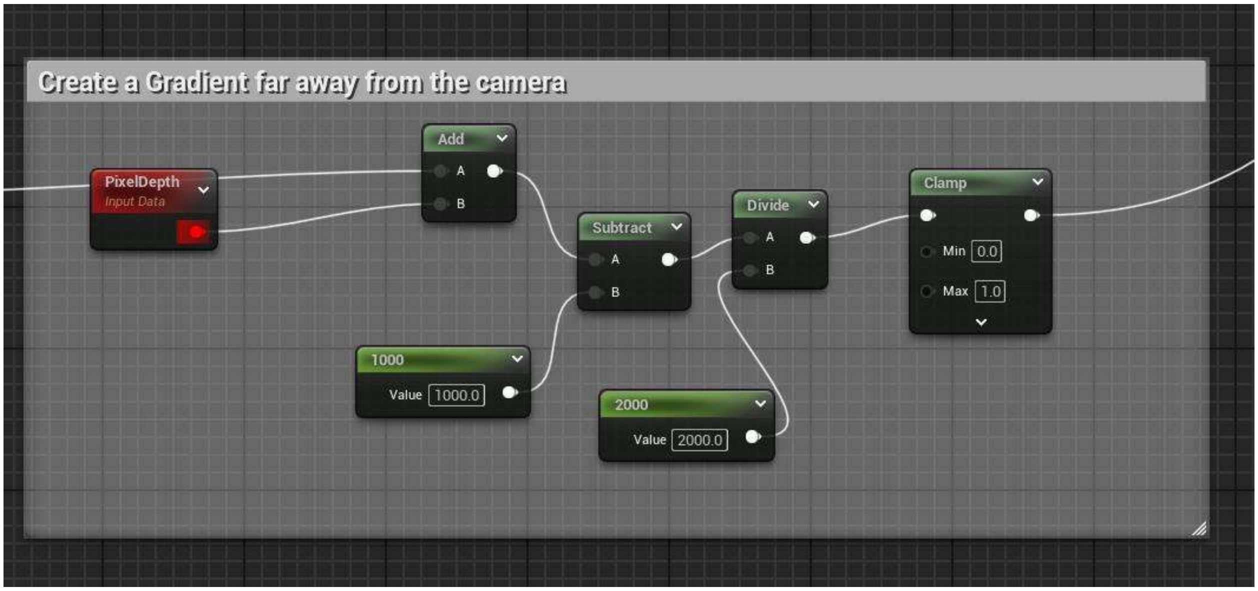

Use Fresnel to Create a Fuzzy Edge Effect
Multiply
A
B
Multiply
A
B
0.5
Value 0.5
Power
Base
Exp
0.4
Value 0.4
Lerp
A
B
Alpha
Fresnel
ExponentIn
BaseReflectFractionIn
0.0
Normal
3
Value 3.0
VertexNormalWS
Input Data


Alpha Of Base Color Defines Roughness
Multiply
A
B
0.9
Value 0.9
0.5
Value 0.5
Lerp
A
B
Alpha
Texture Sample
UVs 0
Tex
Apply View MipBias
RGB
R
G
B
A
RGBA
Texture Sample
UVs 0
Tex
Apply View MipBias
RGB
R
G
B
A
RGBA

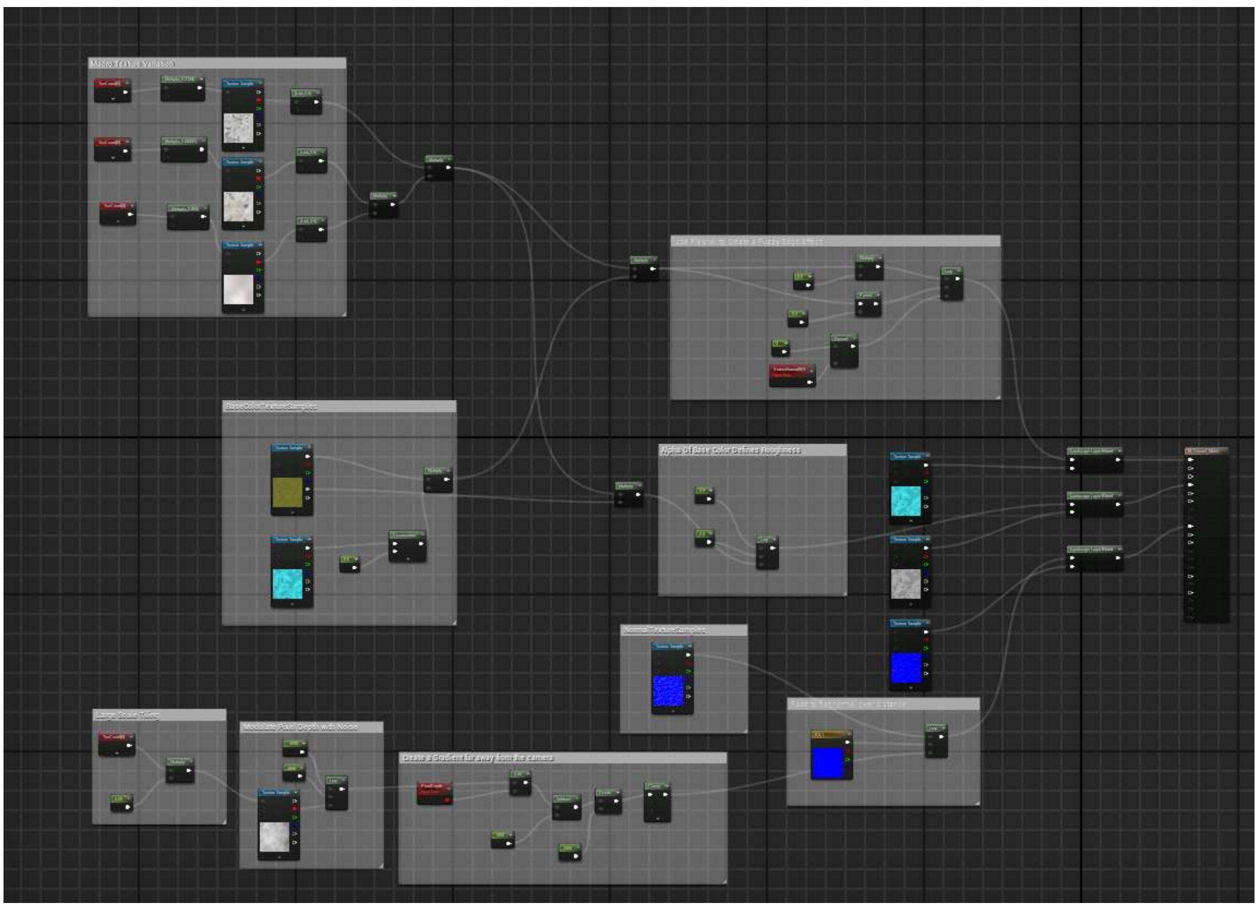

Texture Sample
UVs 0
Tex
Apply View MipBias
RGB
R
G
B
A
RGBA
Landscape Layer Blend
Layer Moss
Layer Path
M_Ground_Moss
Base Color
Metallic 0.0
Specular 0.2
Roughness
Anisotropy 0.0
Emissive Color
Opacity
Opacity Mask
Normal
Tangent X 1.0 Y 0.0 Z 0.0
World Position Offset X 0.0 Y 0.0 Z 0.0
Subsurface Color
Custom Data 0
Custom Data 1
Ambient Occlusion 0.5
Refraction (Disabled)
Pixel Depth Offset 0.0
Shading Model
Surface Thickness
Front Material


NormalTextureSamples
Texture Sample
UVs 0
Tex
Apply View MipBias
RGB
R
G
B
A
RGBA
Fade to flat normal over distance
0,0,1
Constant
X 0.0 Y 0.0 Z 1.0
Lerp
A
B
Alpha

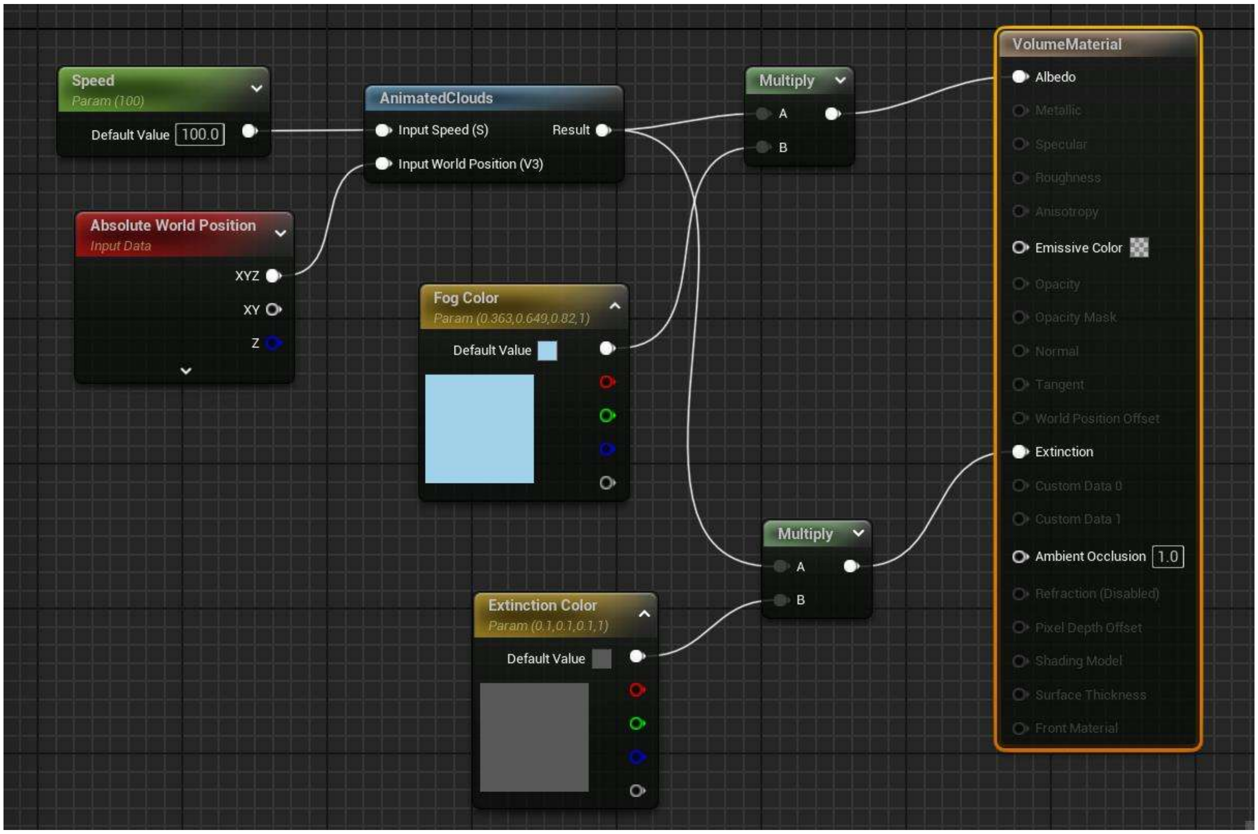

Large Scale Tiling
TexCoord[0]
Coordinate Index 0
Multiply
A
B
0.05
Value 0.05
Modulate Pixel Depth with Noise
-2000
Value -2000.0
2000
Value 2000.0
Lerp
A
B
Alpha
Texture Sample
UVs
Tex
Apply View MipBias
RGB
R
G
B
A
RGBA



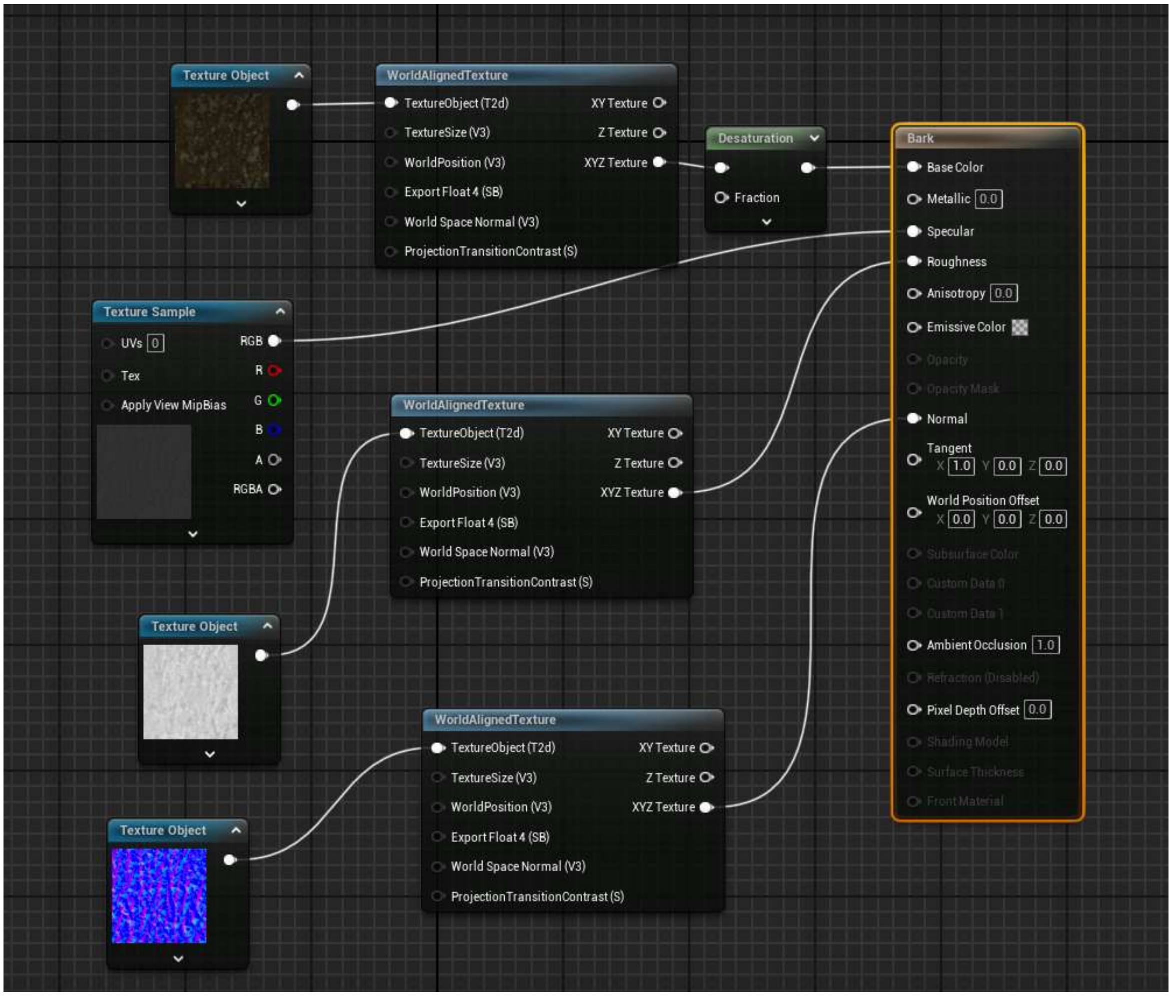
All Together



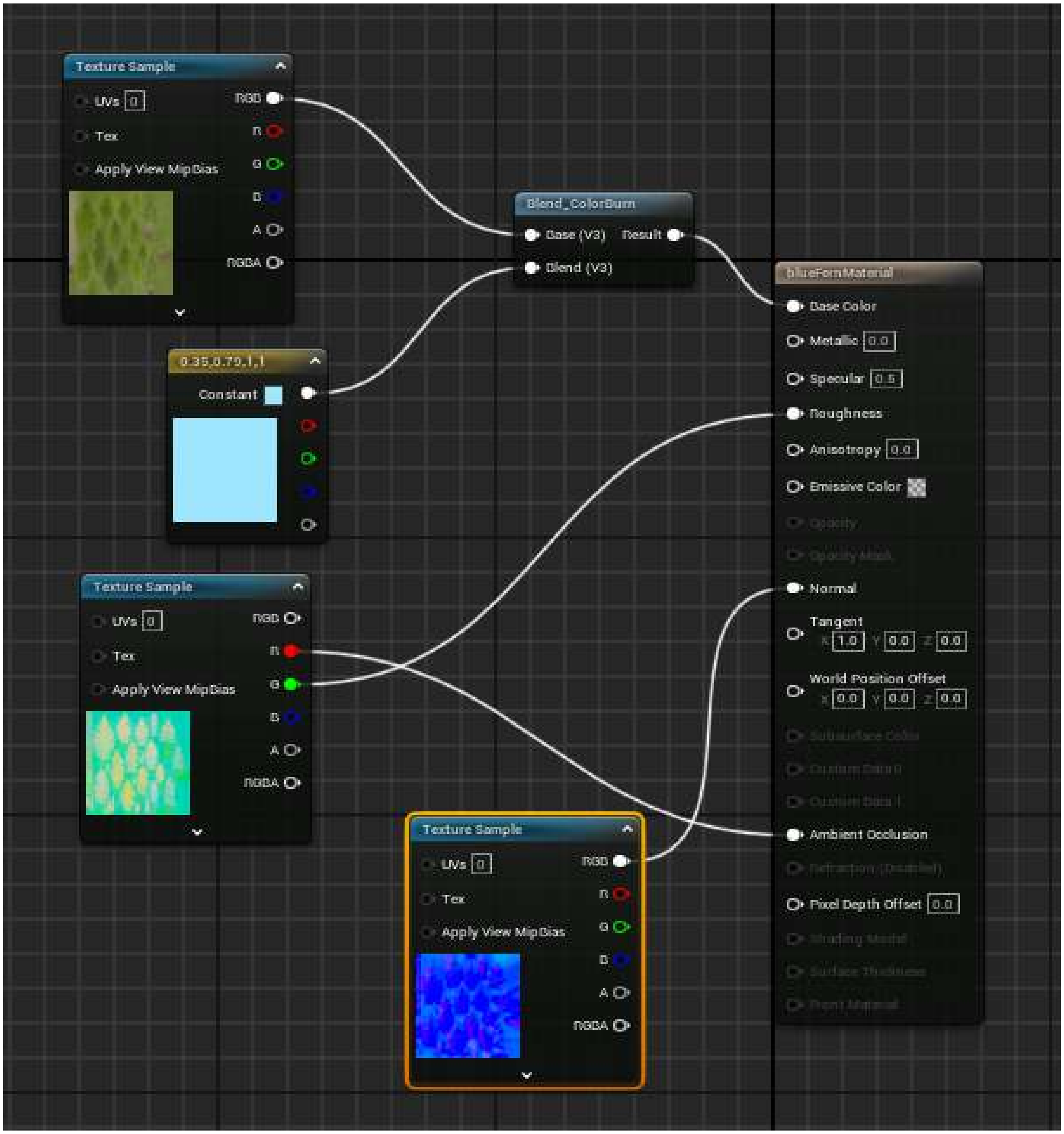
Fog - Volumetric Material



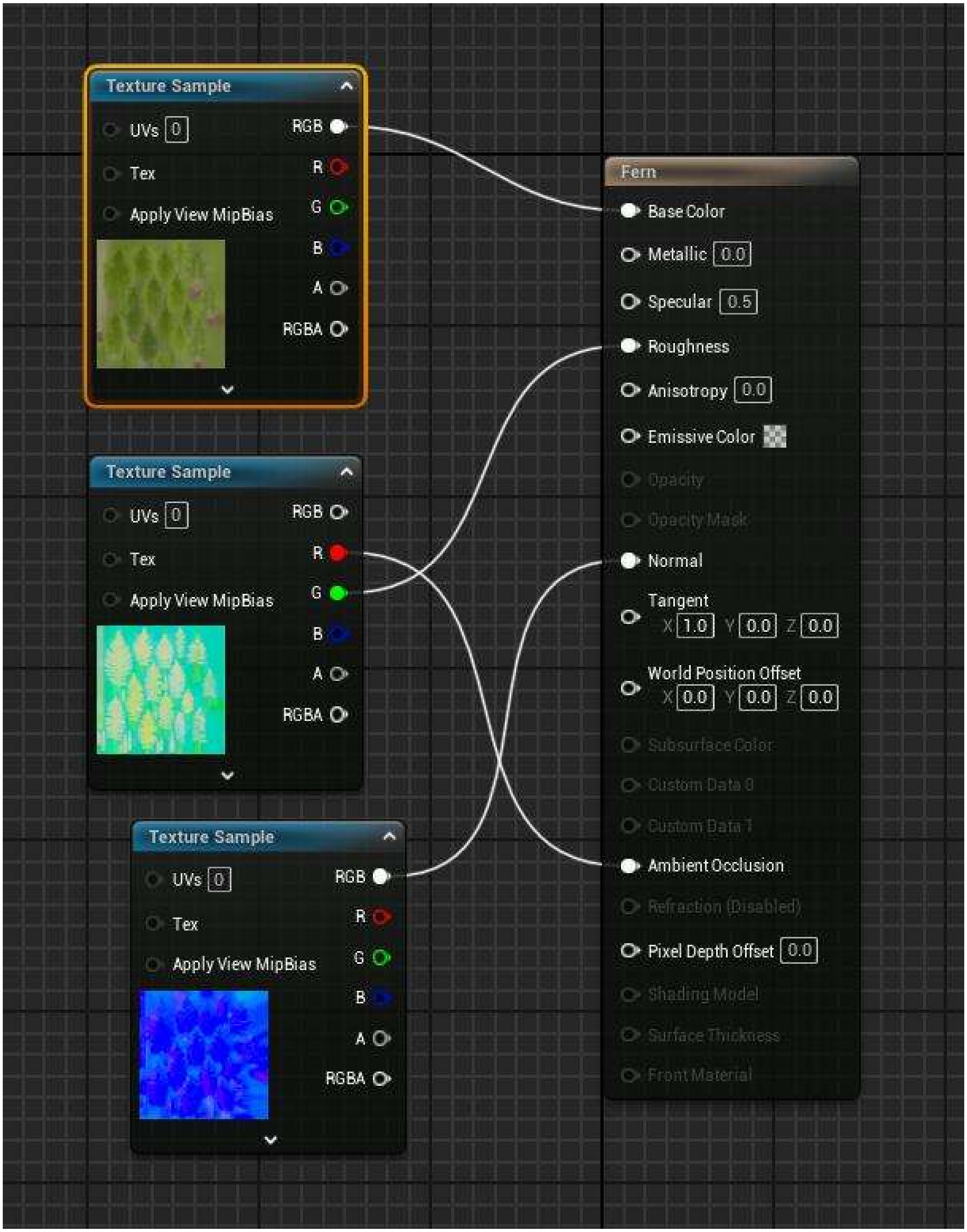
Tree Bark - Altered from a fab lab import



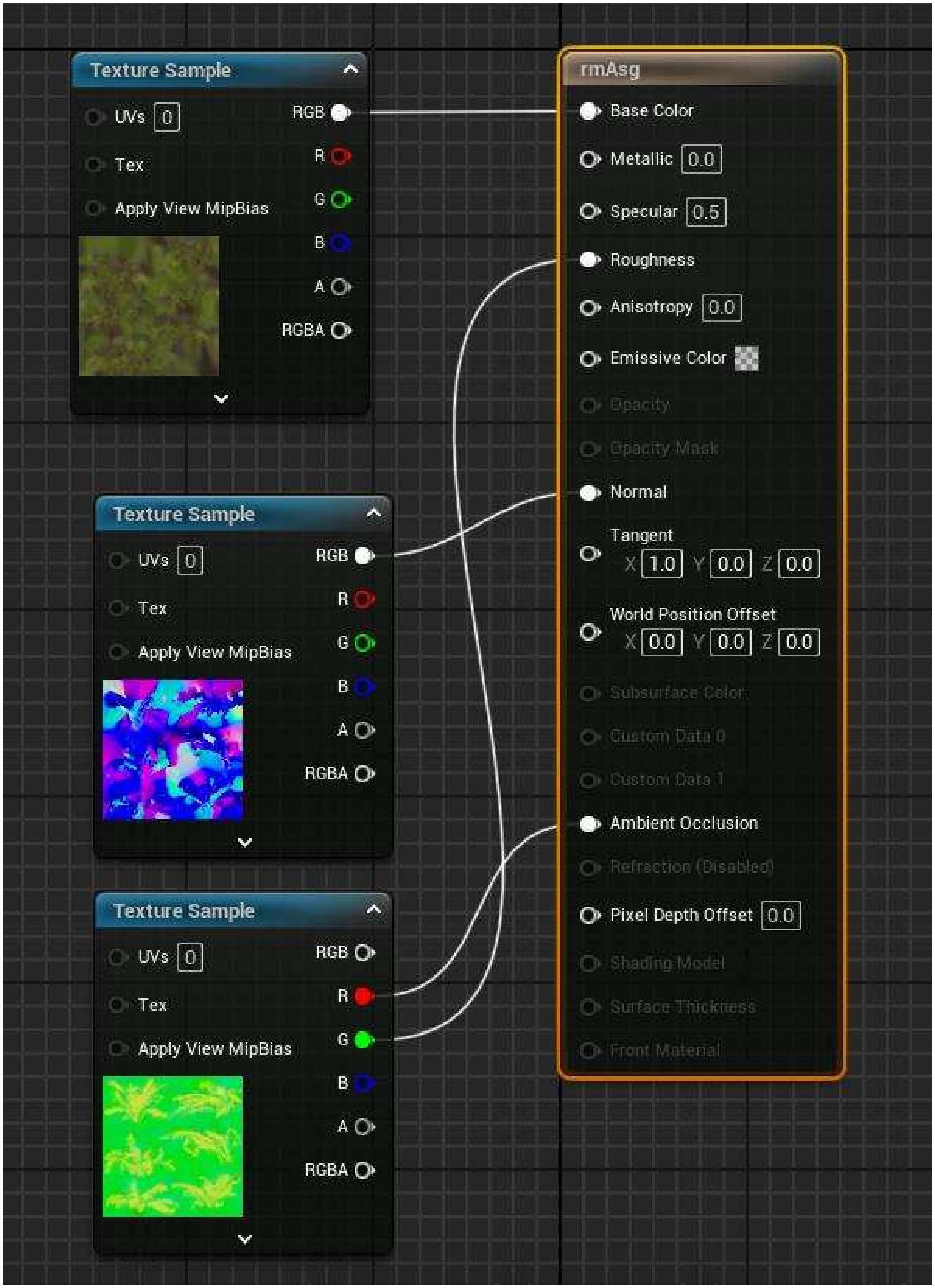
Beech Fern Blue – Altered Colors



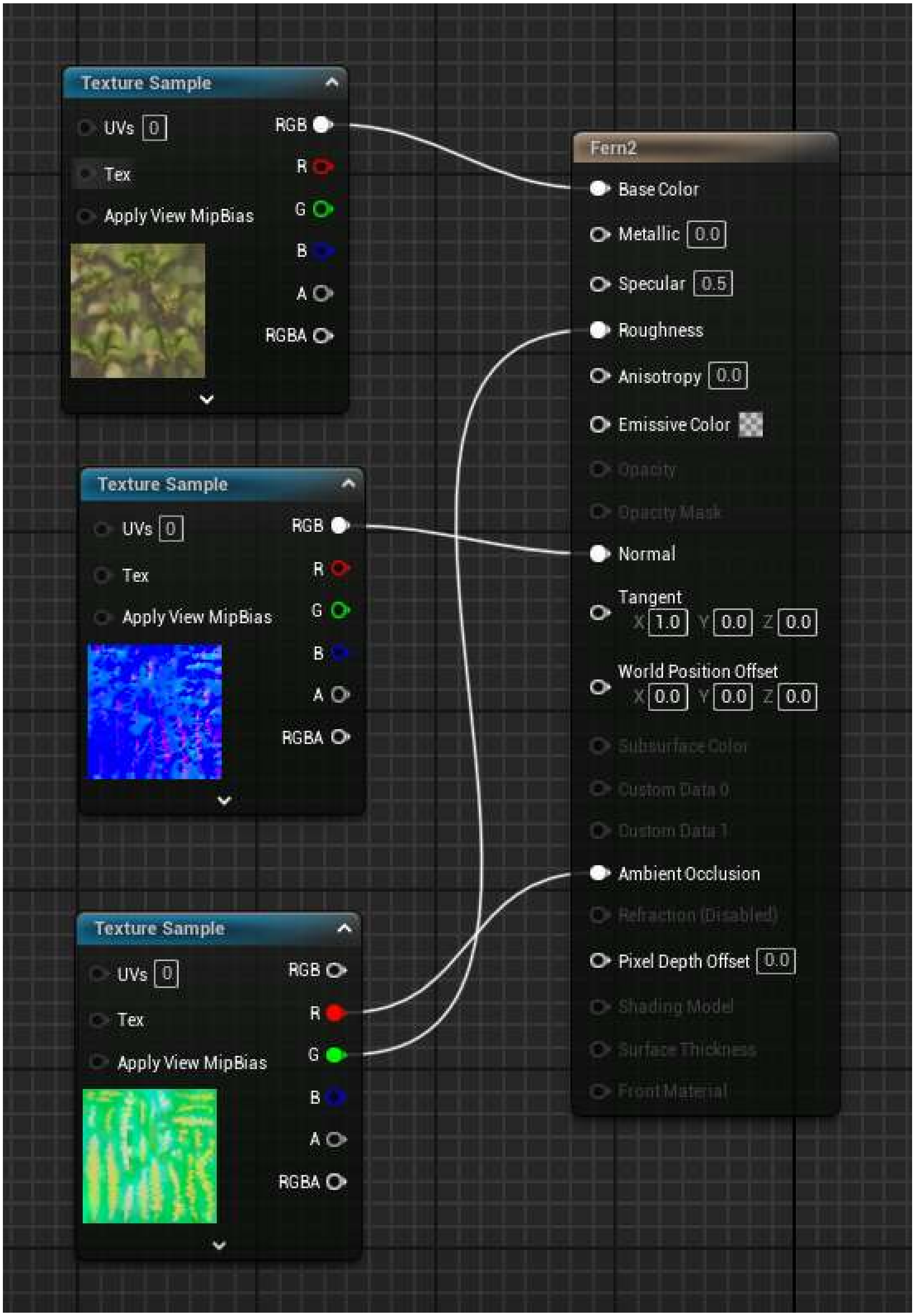
Beech Fern Normal – Created from imported textures



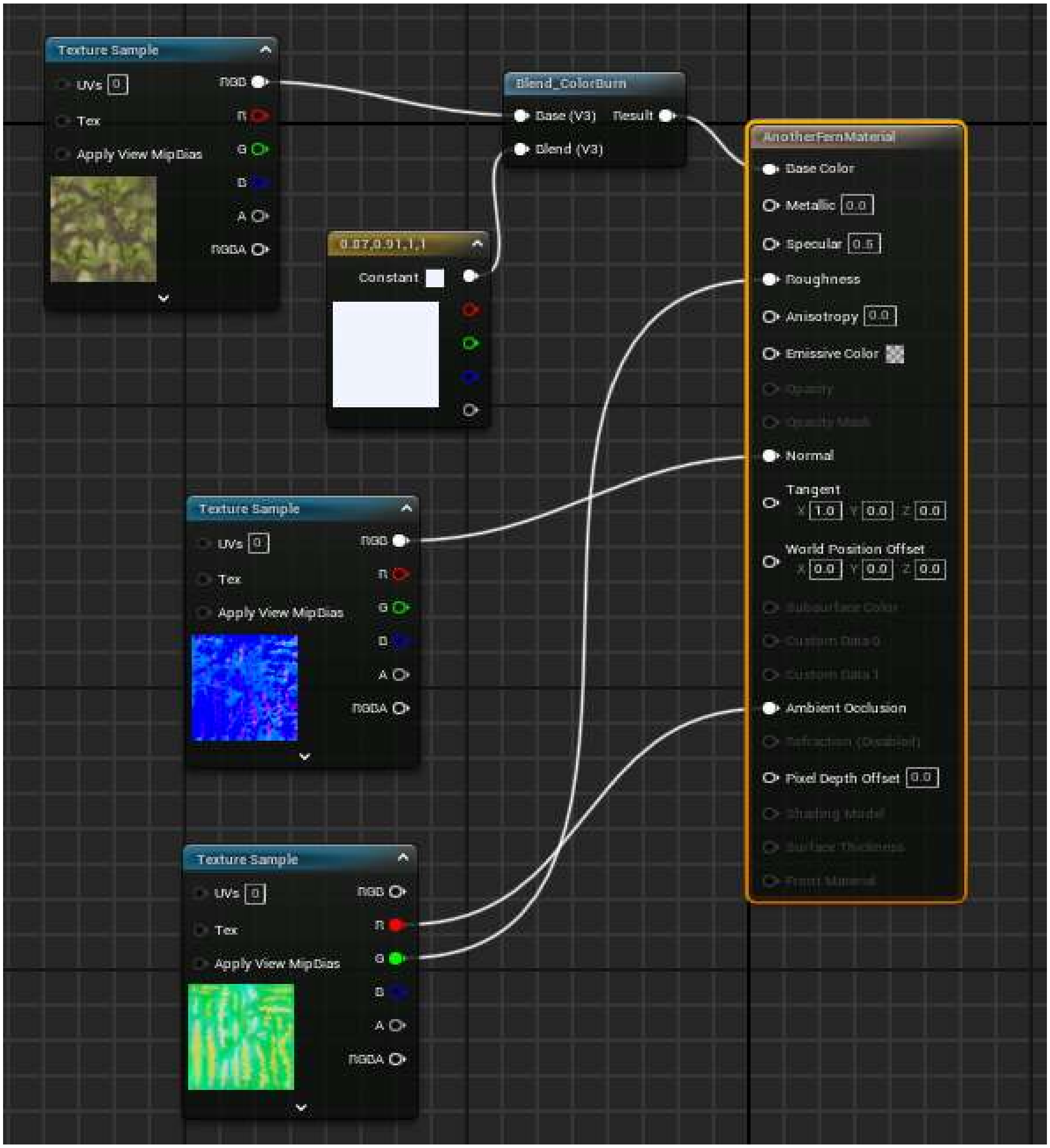
rmAsg Fern – Created from imported textures



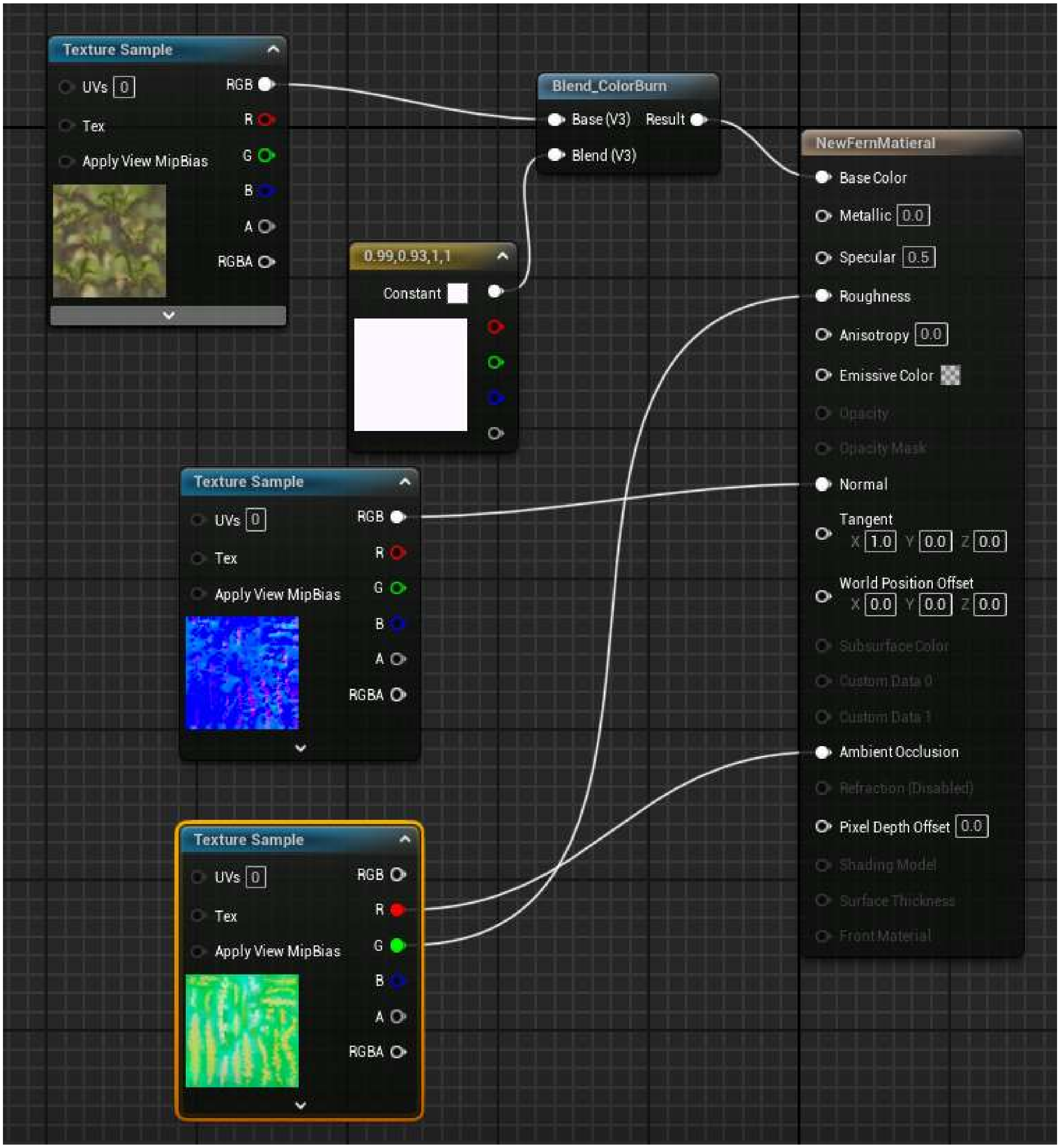
Sbsly Fern - Created from imported textures



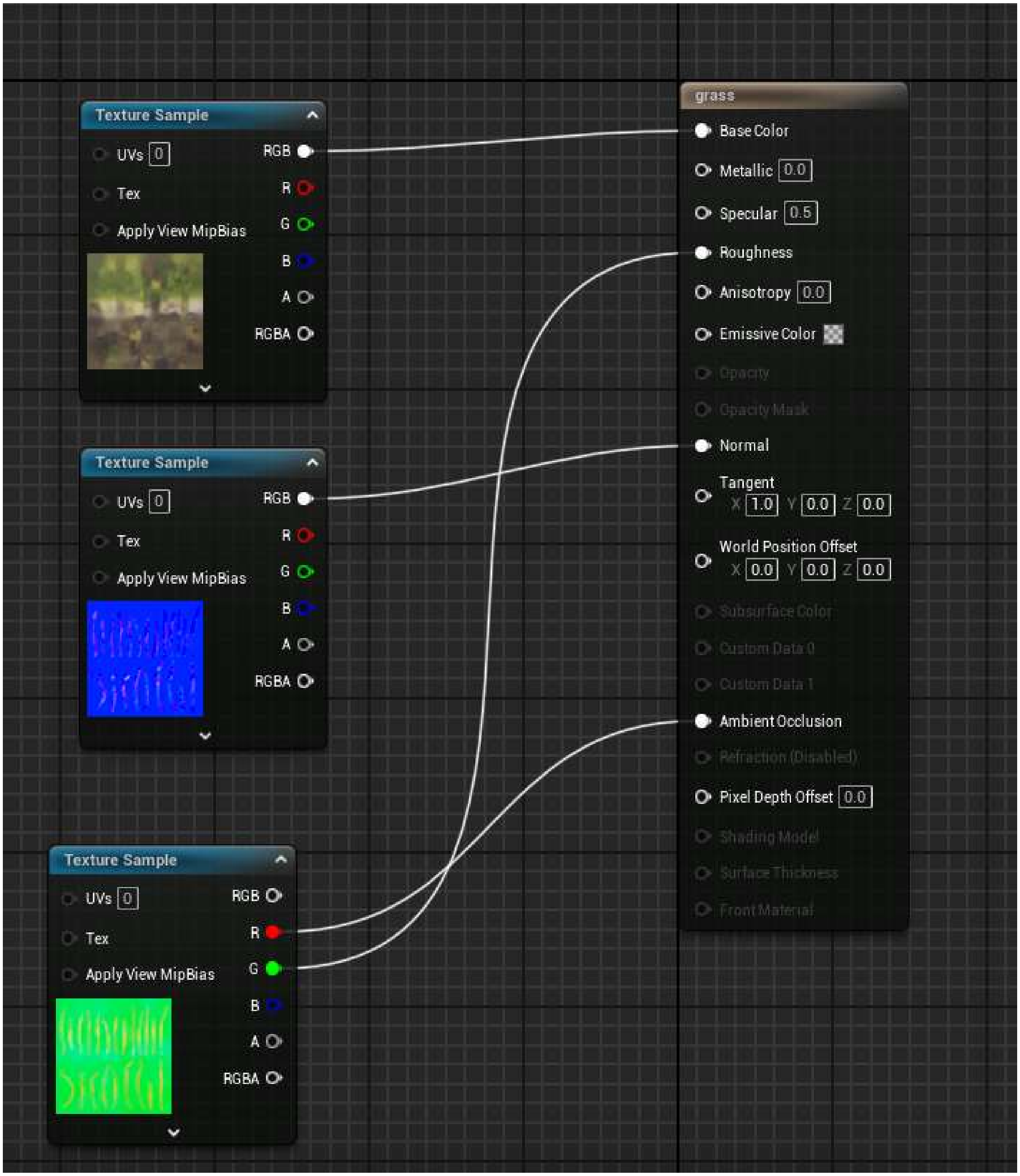
Sbsly Fern Blue and Green - Altered from imported textures



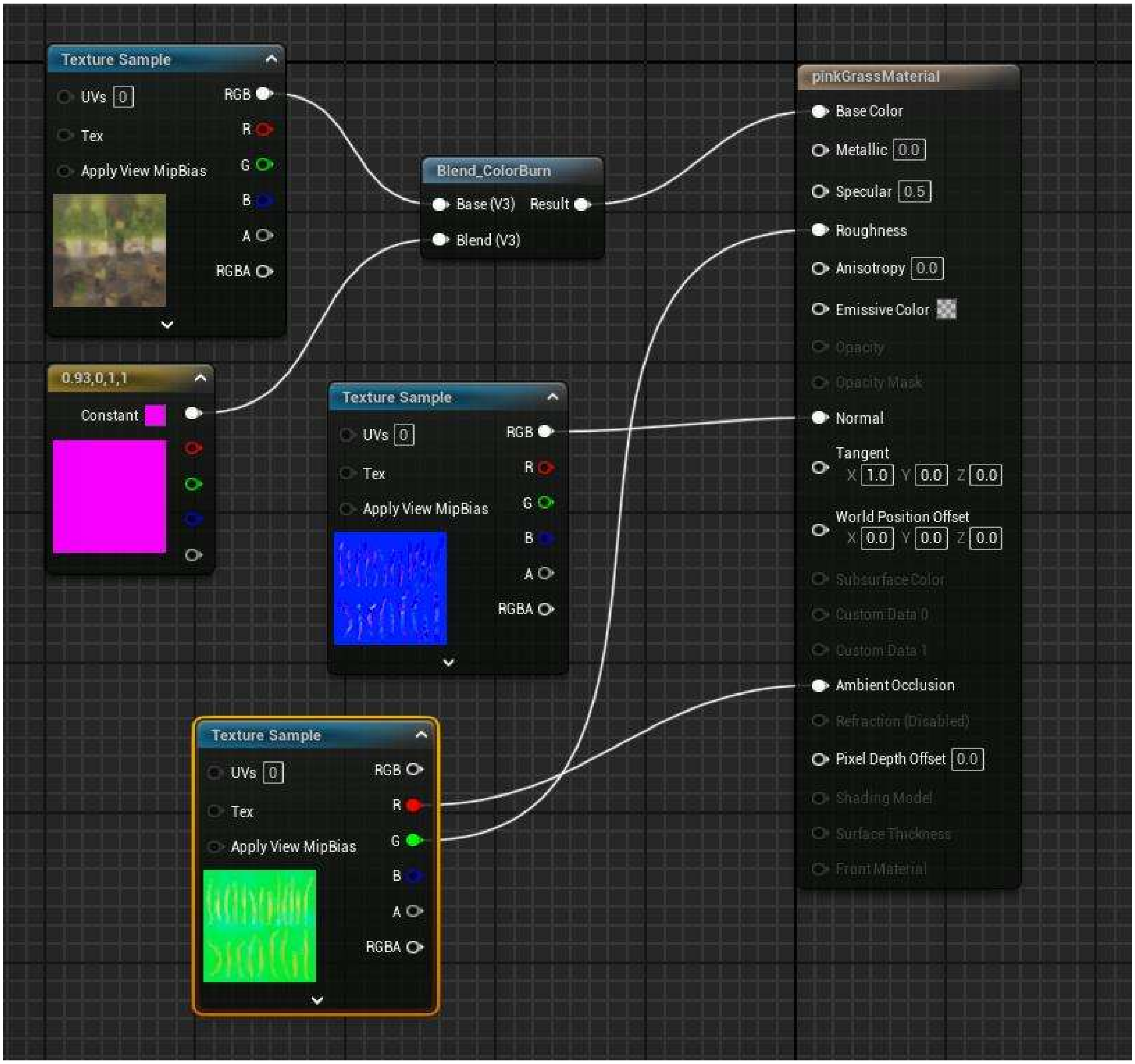
Sbsly Fern Pink and Green - Altered from imported textures



Grass – Created from imported textures

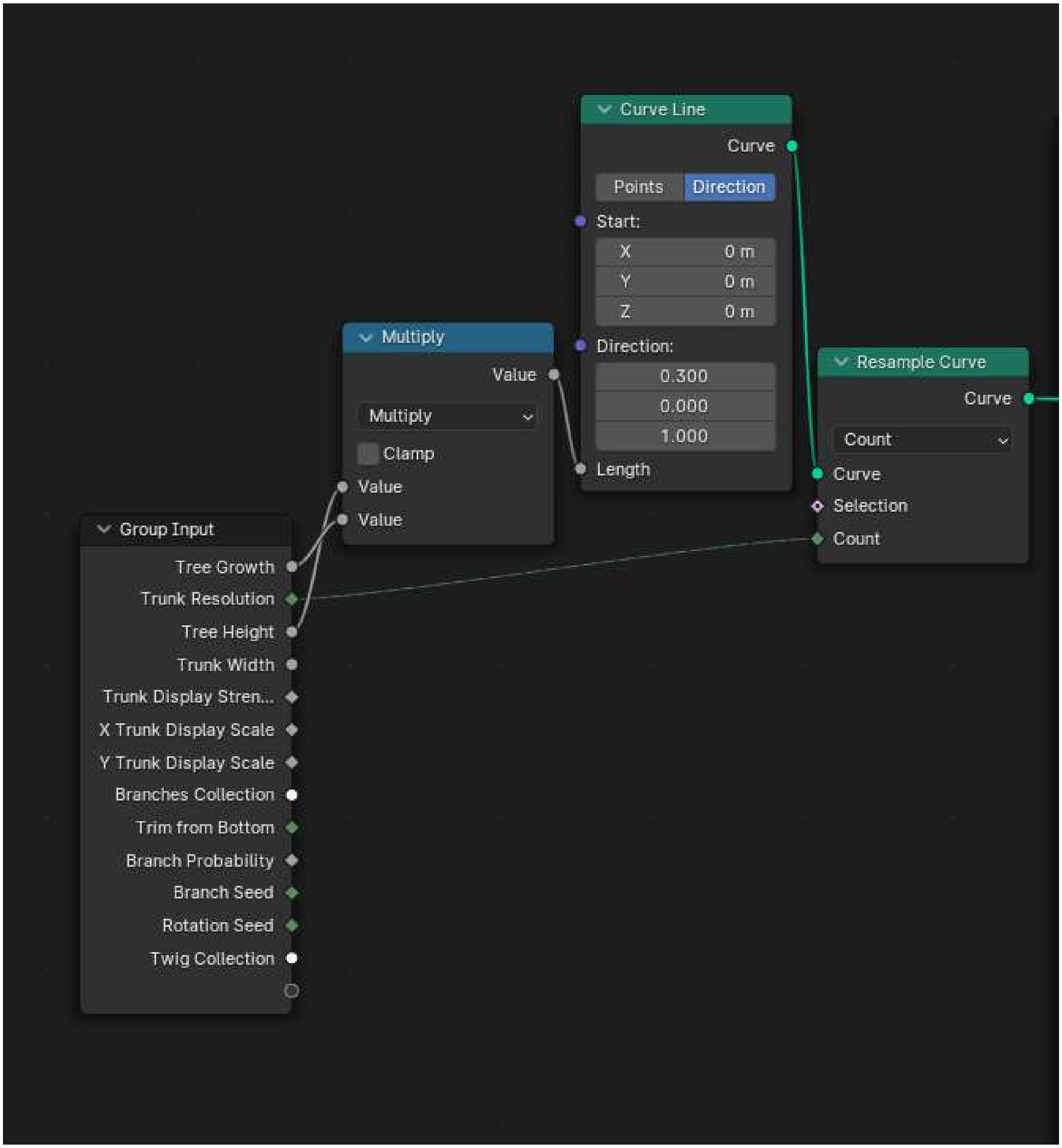


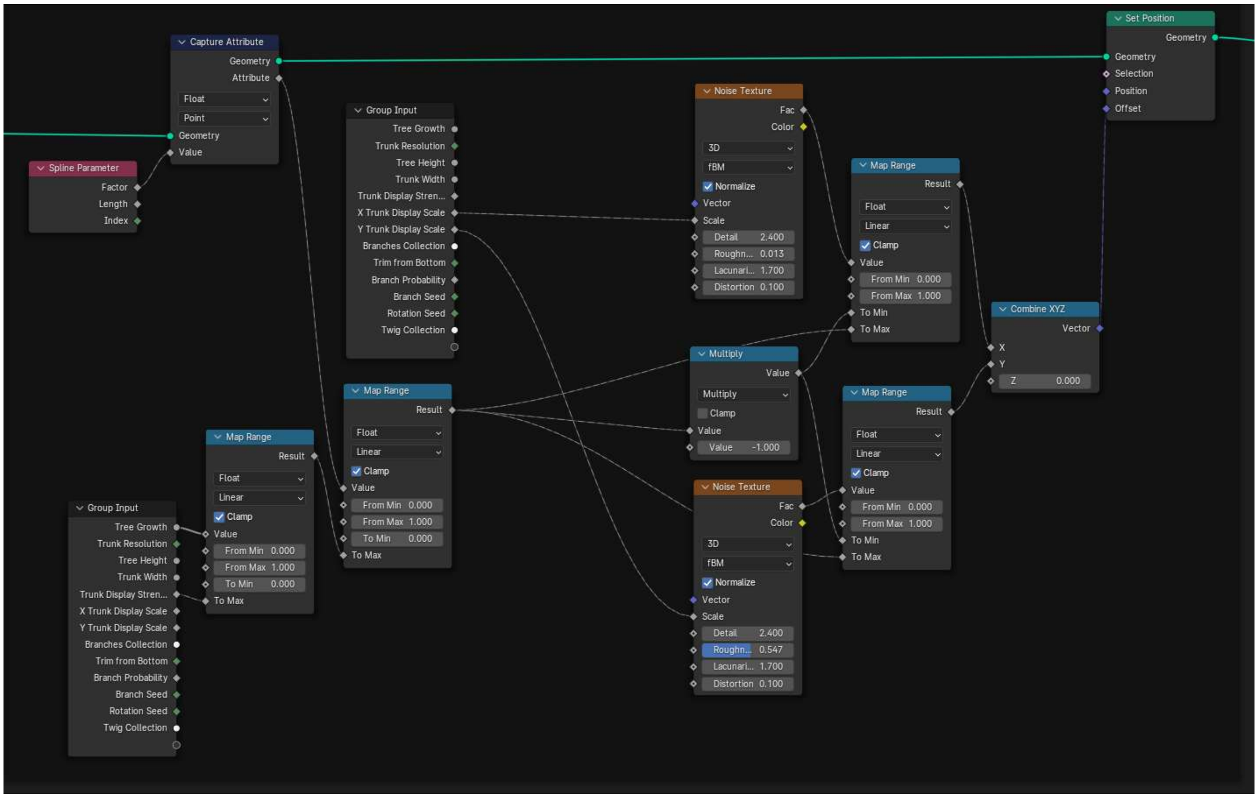
Pink Grass - Altered from imported textures

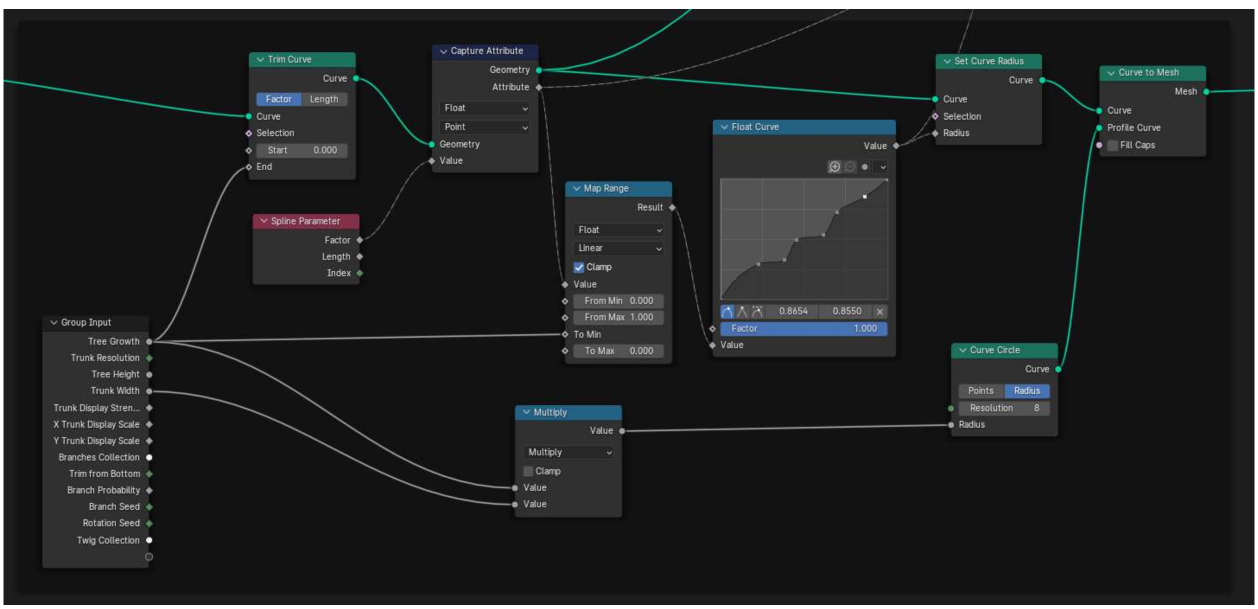


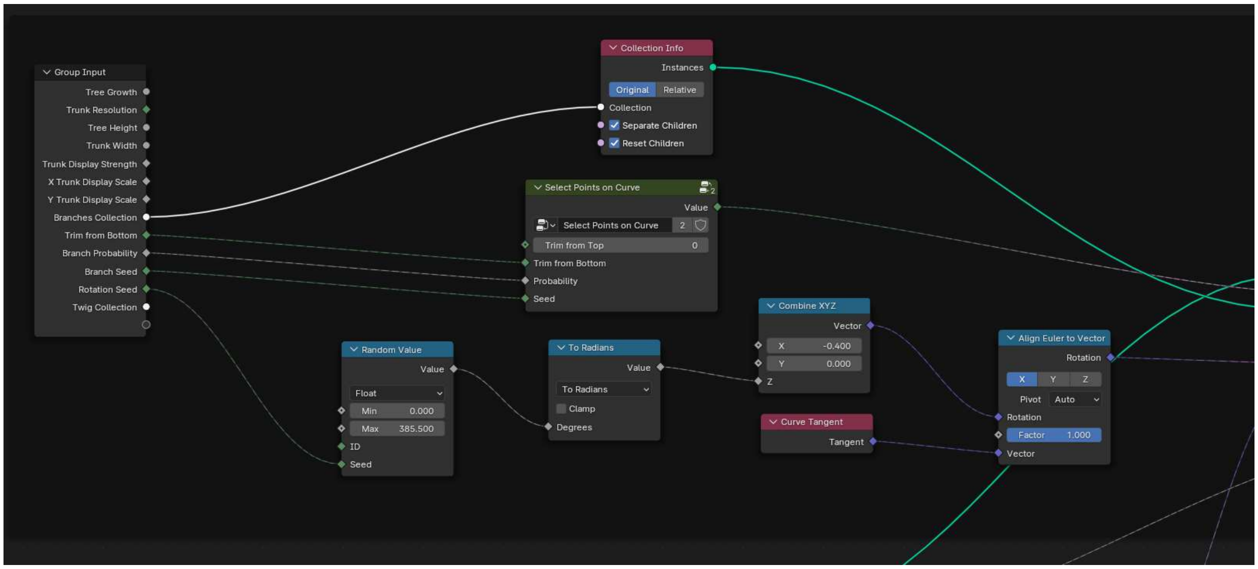
Blender Geometry Nodes

Procedural Tree




Capture Attribute
Geometry
Attribute
Float
Point
Geometry
Value
Spline Parameter
Factor
Length
Index
Group Input
Tree Growth
Trunk Resolution
Tree Height
Trunk Width
Trunk Display Stren...
X Trunk Display Scale
Y Trunk Display Scale
Branches Collection
Trim from Bottom
Branch Probability
Branch Seed
Rotation Seed
Twig Collection
Noise Texture
Fac
Color
3D
fBM
Normalize
Vector
Scale
Detail 2.400
Roughn... 0.013
Lacunari... 1.700
Distortion 0.100
Map Range
Result
Float
Linear
Clamp
Value
From Min 0.000
From Max 1.000
To Min 0.000
To Max
Multiply
Value
Multiply
Clamp
Value -1.000
Noise Texture
Fac
Color
3D
fBM
Normalize
Vector
Scale
Detail 2.400
Roughn... 0.547
Lacunari... 1.700
Distortion 0.100
Combine XYZ
Vector
X
Y
Z 0.000
Set Position
Geometry
Selection
Position
Offset


Trim Curve
Curve
Factor
Length
Curve
Selection
Start 0.000
End
Capture Attribute
Geometry
Attribute
Float
Point
Geometry
Value
Spline Parameter
Factor
Length
Index
Map Range
Result
Float
Linear
Clamp
Value
From Min 0.000
From Max 1.000
To Min
To Max 0.000
Float Curve
Value
0.8654
0.8550
Factor 1.000
Value
Set Curve Radius
Curve
Curve
Selection
Radius
Curve to Mesh
Mesh
Curve
Profile Curve
Fill Caps
Curve Circle
Curve
Points
Radius
Resolution 8
Radius
Group Input
Tree Growth
Trunk Resolution
Tree Height
Trunk Width
Trunk Display Stren...
X Trunk Display Scale
Y Trunk Display Scale
Branches Collection
Trim from Bottom
Branch Probability
Branch Seed
Rotation Seed
Twig Collection
Multiply
Value
Multiply
Clamp
Value
Value


Group Input
Tree Growth
Trunk Resolution
Tree Height
Trunk Width
Trunk Display Strength
X Trunk Display Scale
Y Trunk Display Scale
Branches Collection
Trim from Bottom
Branch Probability
Branch Seed
Rotation Seed
Twig Collection
Collection Info
Instances
Original
Relative
Collection
Separate Children
Reset Children
Select Points on Curve
Value
Select Points on Curve
Trim from Top
Trim from Bottom
Probability
Seed
Random Value
Value
Float
Min 0.000
Max 385.500
ID
Seed
To Radians
Value
To Radians
Clamp
Degrees
Combine XYZ
Vector
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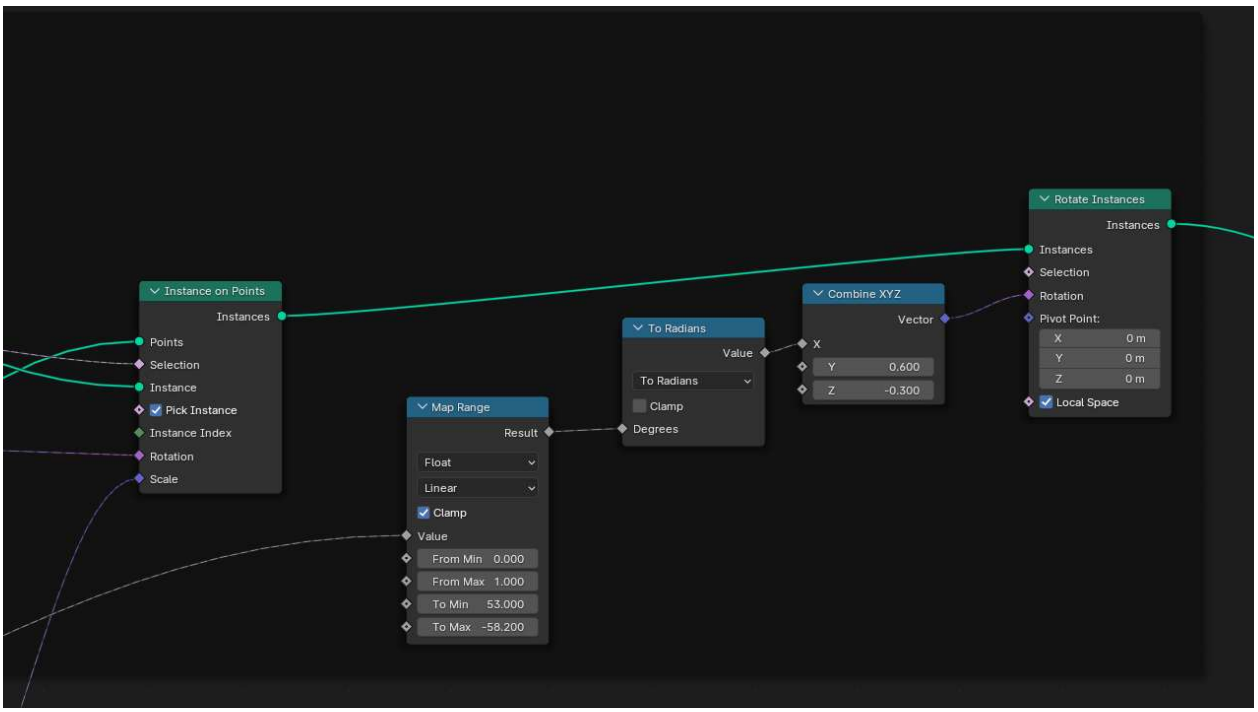
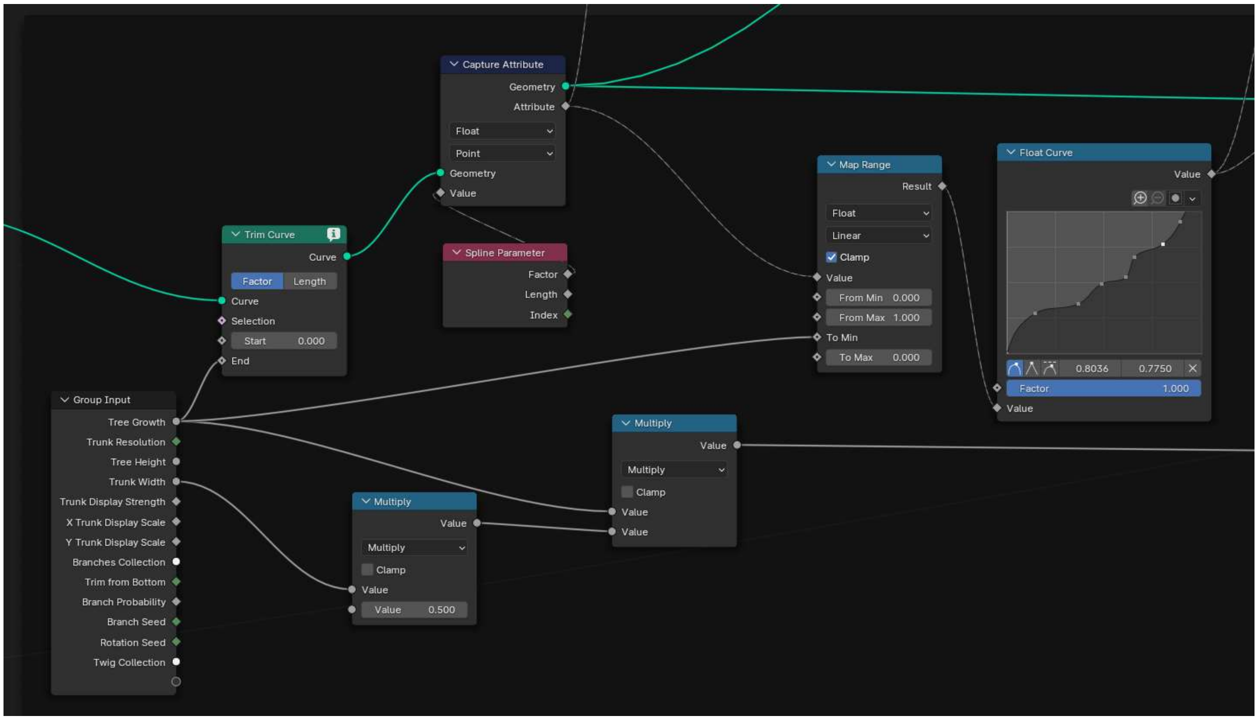
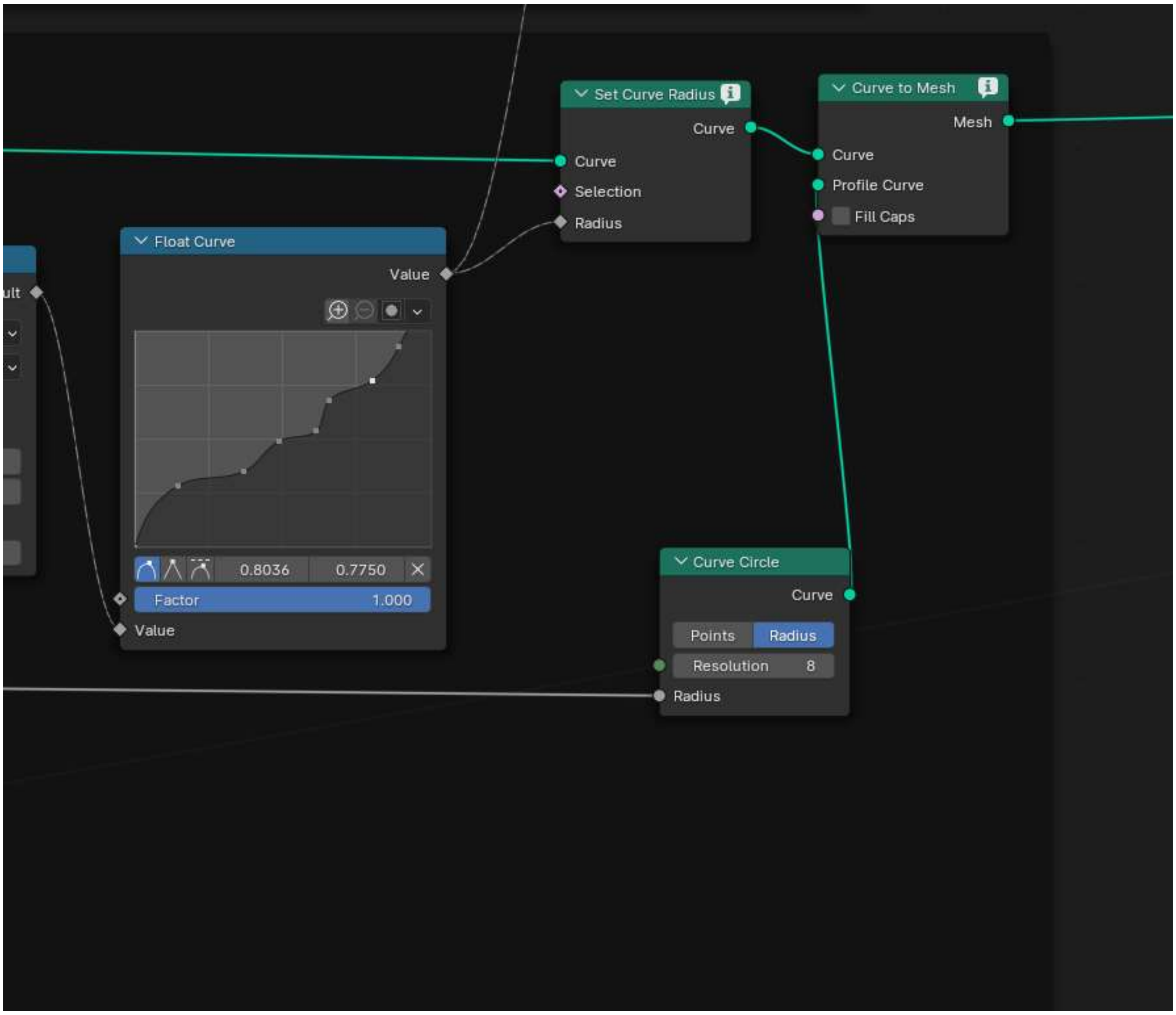
Y 0.000
Z
Curve Tangent
Tangent
Align Euler to Vector
Rotation
X Y Z
Pivot Auto
Rotation
Factor 1.000
Vector


Instance on Points
Instances
Points
Selection
Instance
Pick Instance
Instance Index
Rotation
Scale
Map Range
Result
Float
Linear
Clamp
Value
From Min 0.000
From Max 1.000
To Min 53.000
To Max -58.200
To Radians
Value
To Radians
Clamp
Degrees
Combine XYZ
Vector
X
Y 0.600
Z -0.300
Rotate Instances
Instances
Instances
Selection
Rotation
Pivot Point:
X 0 m
Y 0 m
Z 0 m
Local Space


Capture Attribute
Geometry
Attribute
Float
Point
Geometry
Value
Map Range
Result
Float
Linear
Clamp
Value
From Min 0.000
From Max 1.000
To Min
To Max 0.000
Float Curve
Value
0.8036
0.7750
Factor 1.000
Value
Trim Curve
Curve
Factor
Length
Curve
Selection
Start 0.000
End
Spline Parameter
Factor
Length
Index
Group Input
Tree Growth
Trunk Resolution
Tree Height
Trunk Width
Trunk Display Strength
X Trunk Display Scale
Y Trunk Display Scale
Branches Collection
Trim from Bottom
Branch Probability
Branch Seed
Rotation Seed
Twig Collection
Multiply
Value
Multiply
Clamp
Value
Value
Multiply
Value
Multiply
Clamp
Value
Value 0.500


Set Curve Radius
Curve
Curve
Selection
Radius
Curve to Mesh
Mesh
Curve
Profile Curve
Fill Caps
Float Curve
Value
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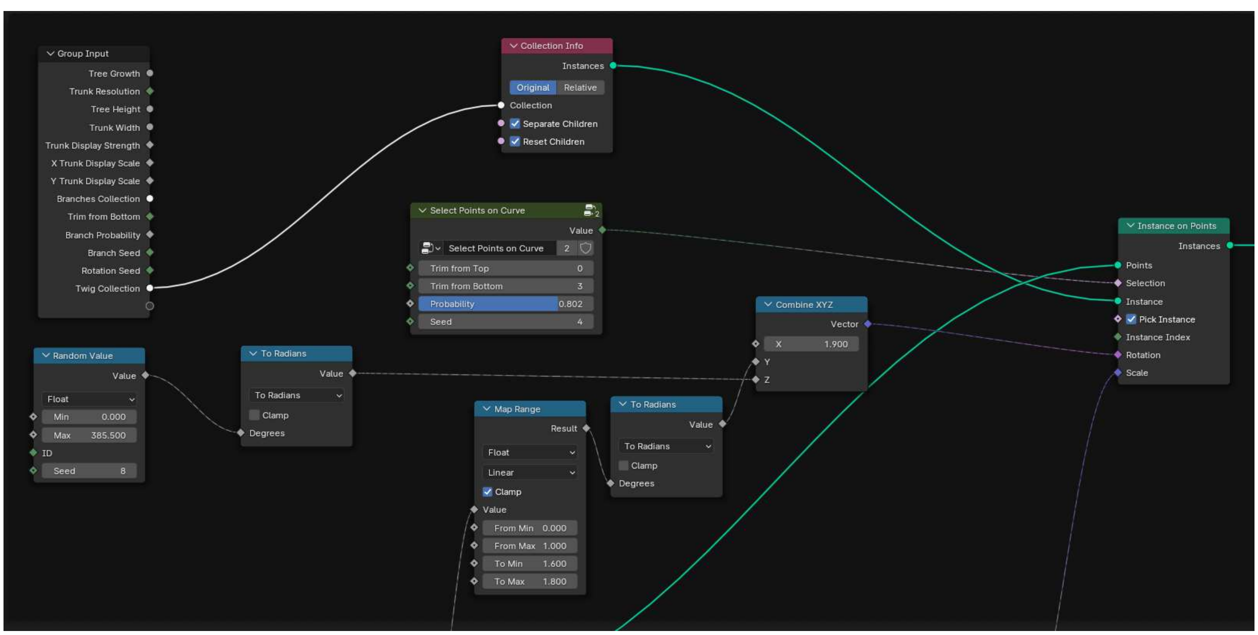
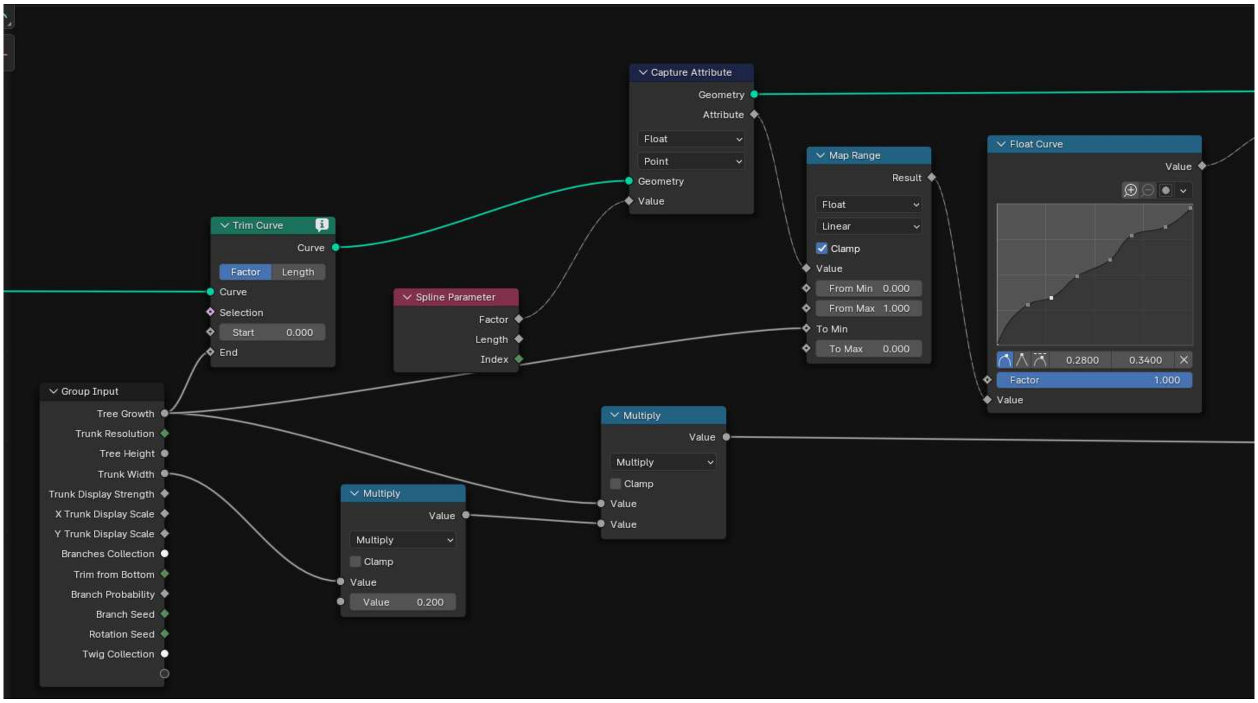
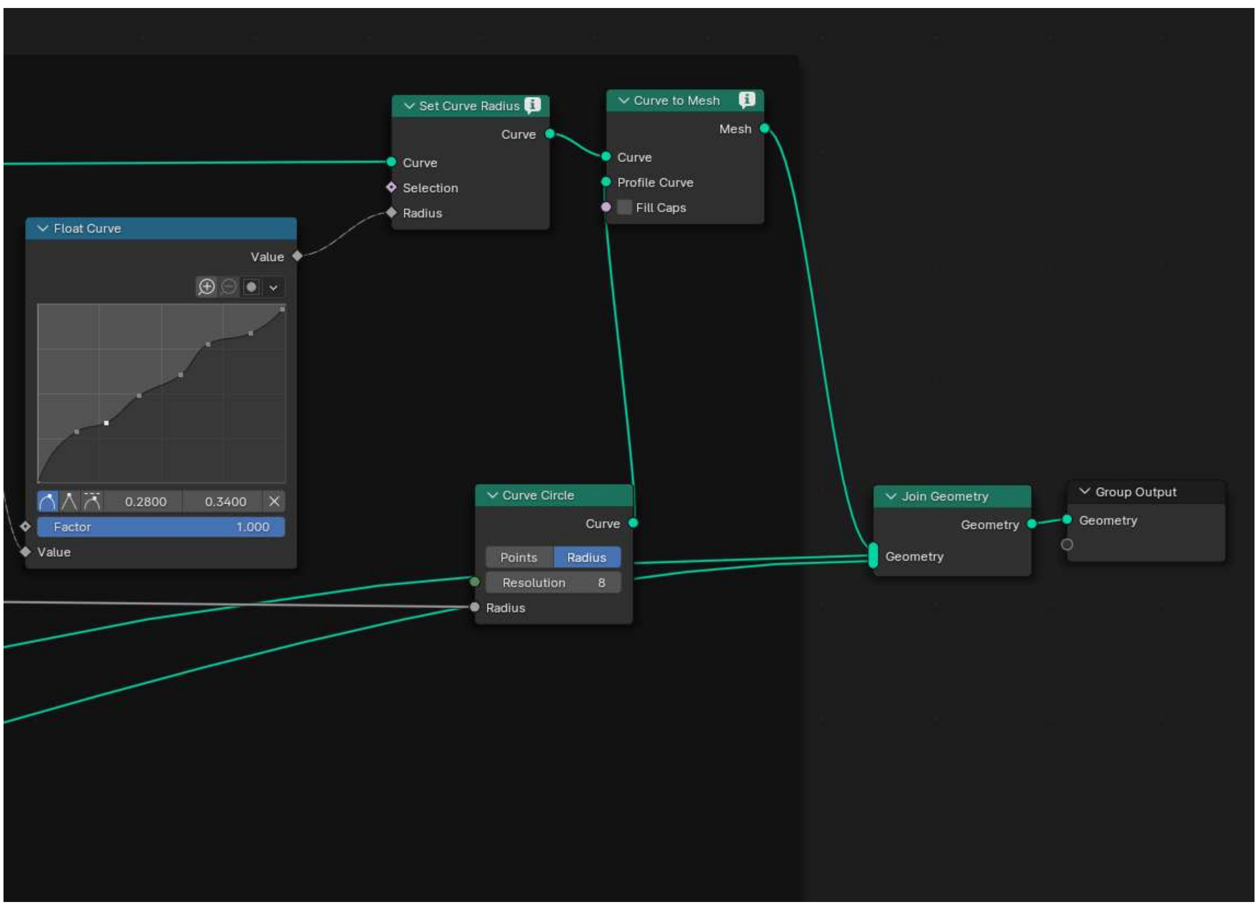
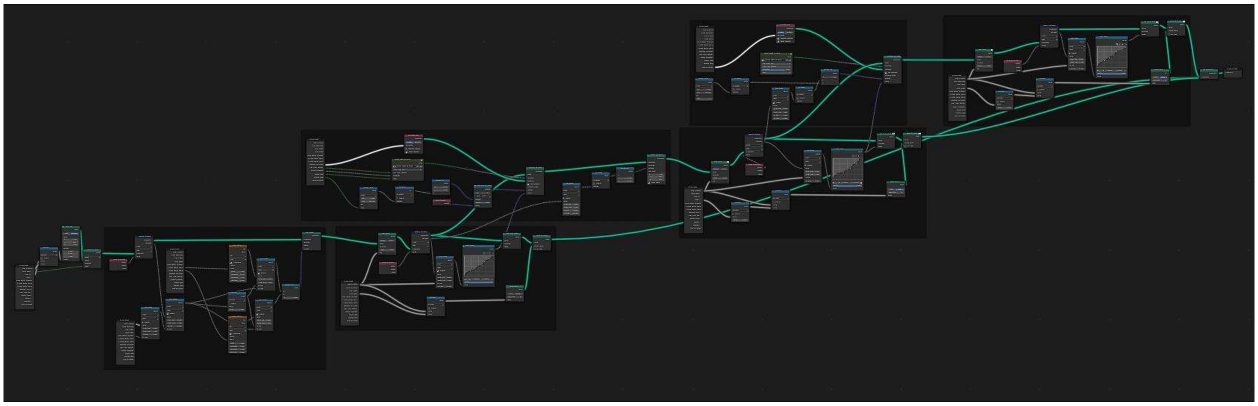
0.7750
Factor
1.000
Value
Curve Circle
Curve
Points
Radius
Resolution
8
Radius


Group Input
Tree Growth
Trunk Resolution
Tree Height
Trunk Width
Trunk Display Strength
X Trunk Display Scale
Y Trunk Display Scale
Branches Collection
Trim from Bottom
Branch Probability
Branch Seed
Rotation Seed
Twig Collection
Collection Info
Instances
Original
Relative
Collection
Separate Children
Reset Children
Select Points on Curve
Value
Select Points on Curve
Trim from Top
Trim from Bottom
Probability
0.802
Seed
Instance on Points
Instances
Points
Selection
Instance
Pick Instance
Instance Index
Rotation
Scale
Combine XYZ
Vector
X
1.900
Y
Z
Random Value
Value
Float
Min
0.000
Max
385.500
ID
Seed
8
To Radians
Value
To Radians
Clamp
Degrees
Map Range
Result
Float
Linear
Clamp
Value
From Min
0.000
From Max
1.000
To Min
1.600
To Max
1.800

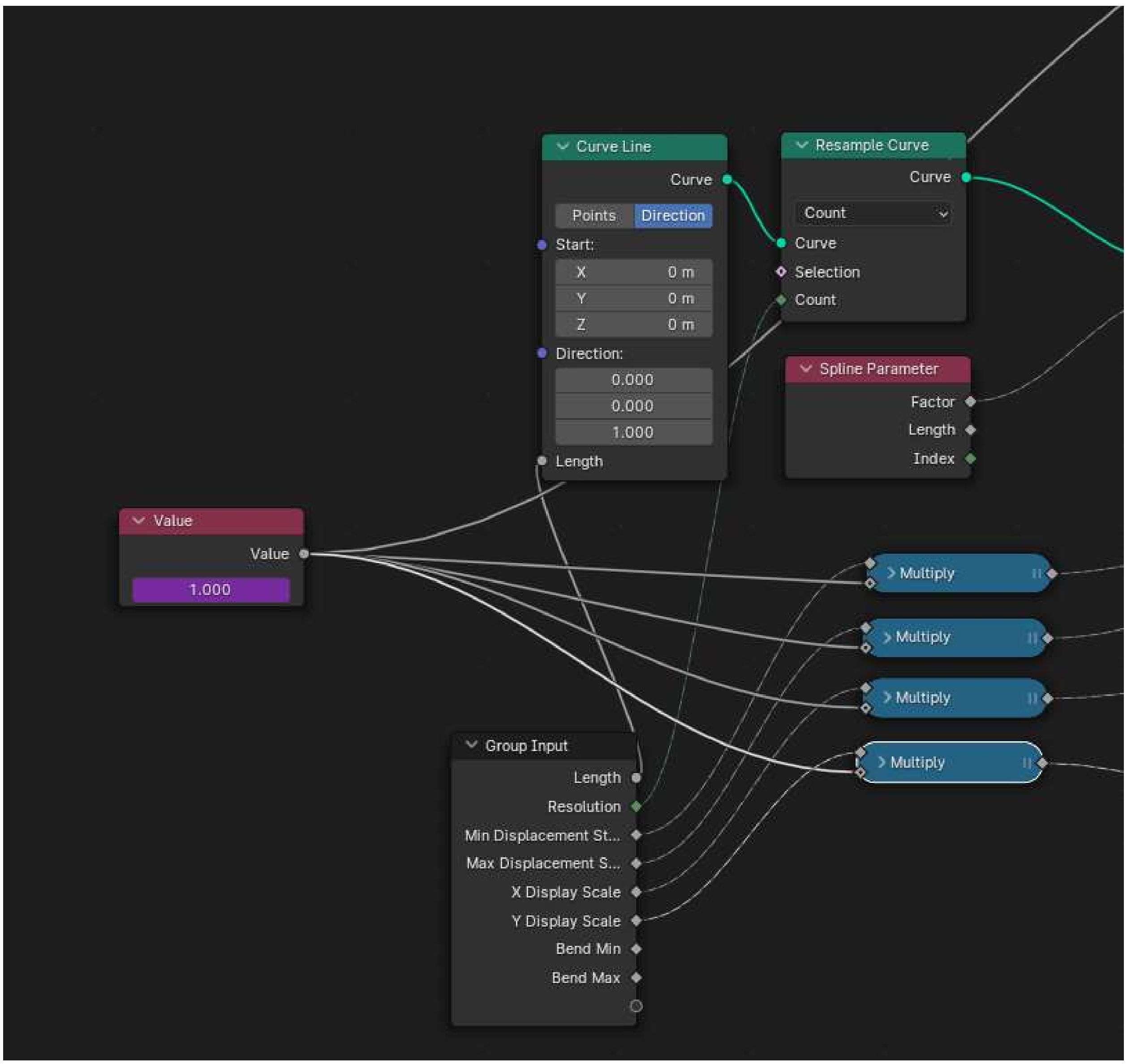

Capture Attribute
Geometry
Attribute
Float
Point
Geometry
Value
Map Range
Result
Float
Linear
Clamp
Value
From Min 0.000
From Max 1.000
To Min
To Max 0.000
Float Curve
Value
0.2800
0.3400
Factor 1.000
Value
Trim Curve
Curve
Factor
Length
Curve
Selection
Start 0.000
End
Spline Parameter
Factor
Length
Index
Group Input
Tree Growth
Trunk Resolution
Tree Height
Trunk Width
Trunk Display Strength
X Trunk Display Scale
Y Trunk Display Scale
Branches Collection
Trim from Bottom
Branch Probability
Branch Seed
Rotation Seed
Twig Collection
Multiply
Value
Multiply
Clamp
Value
Value
Multiply
Value
Multiply
Clamp
Value
Value 0.200

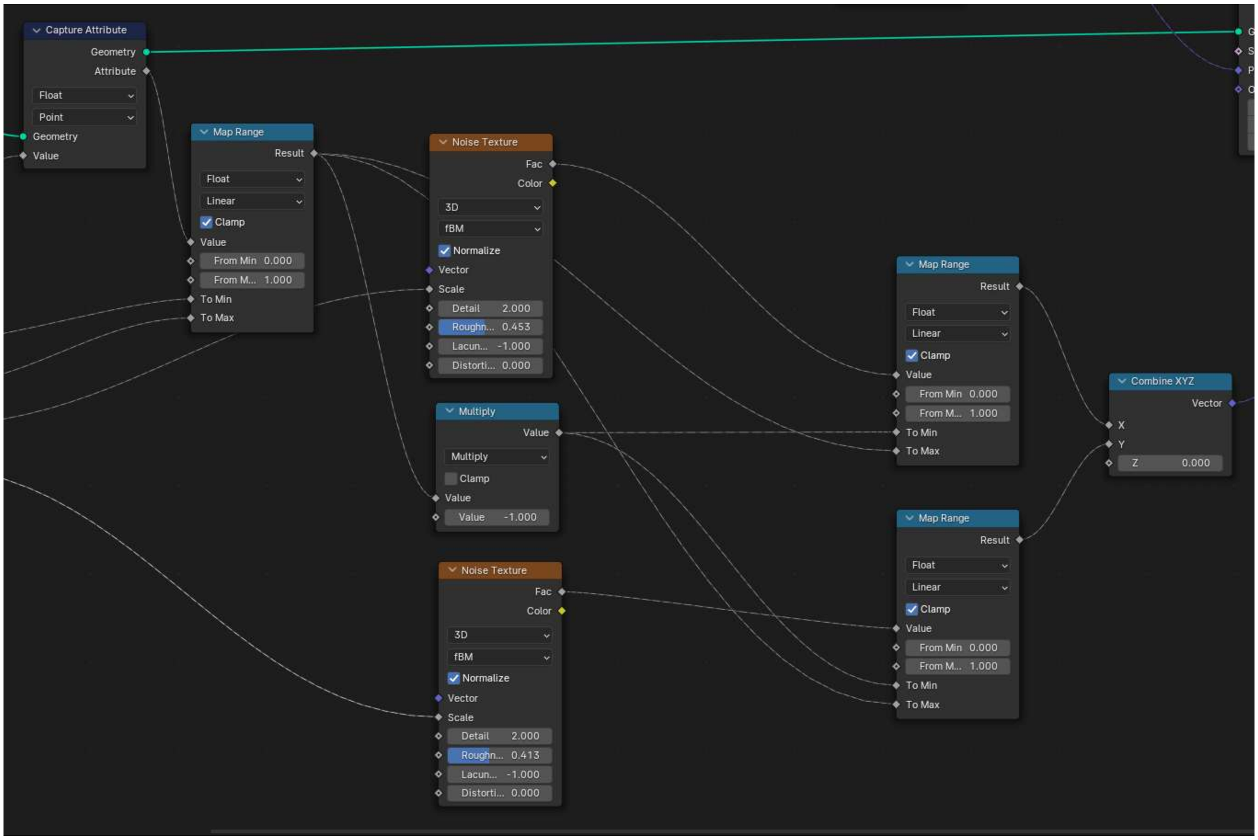

Set Curve Radius
Curve
Curve
Selection
Radius
Curve to Mesh
Mesh
Curve
Profile Curve
Fill Caps
Float Curve
Value
0.2800
0.3400
Factor 1.000
Value
Curve Circle
Curve
Points
Radius
Resolution 8
Radius
Join Geometry
Geometry
Geometry
Group Output
Geometry

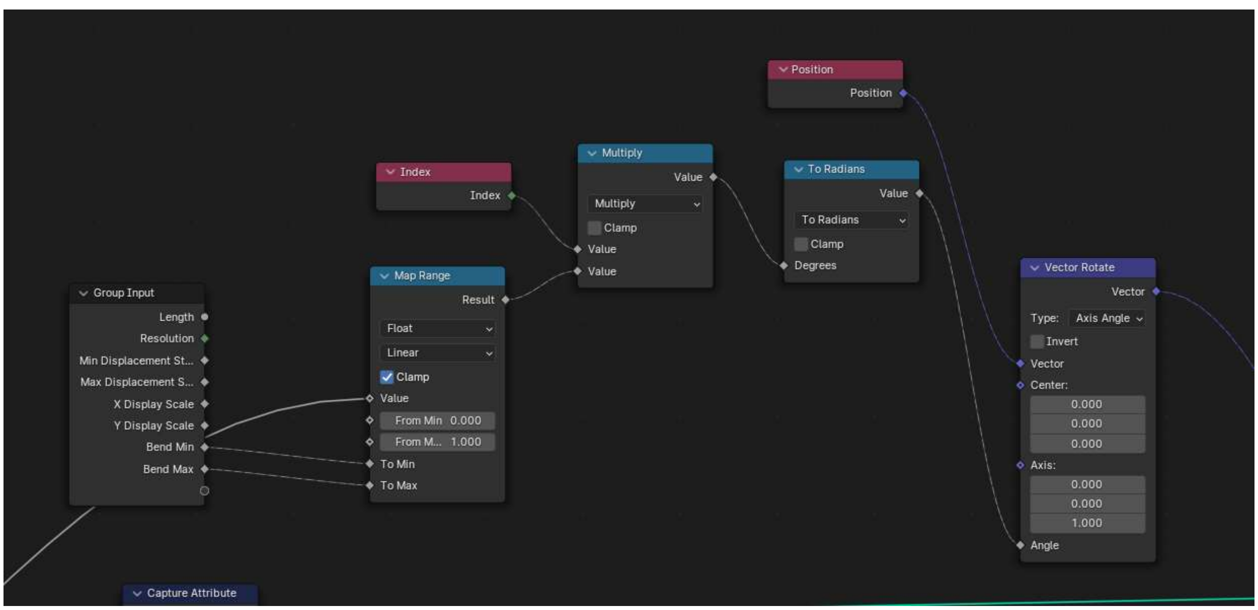
All Together

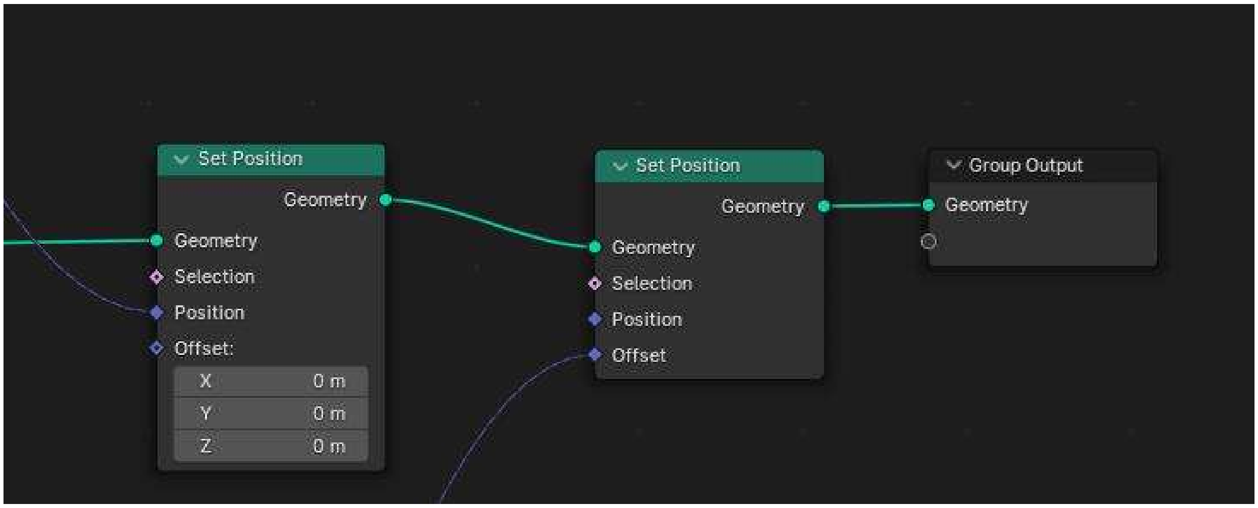


Procedural Branches

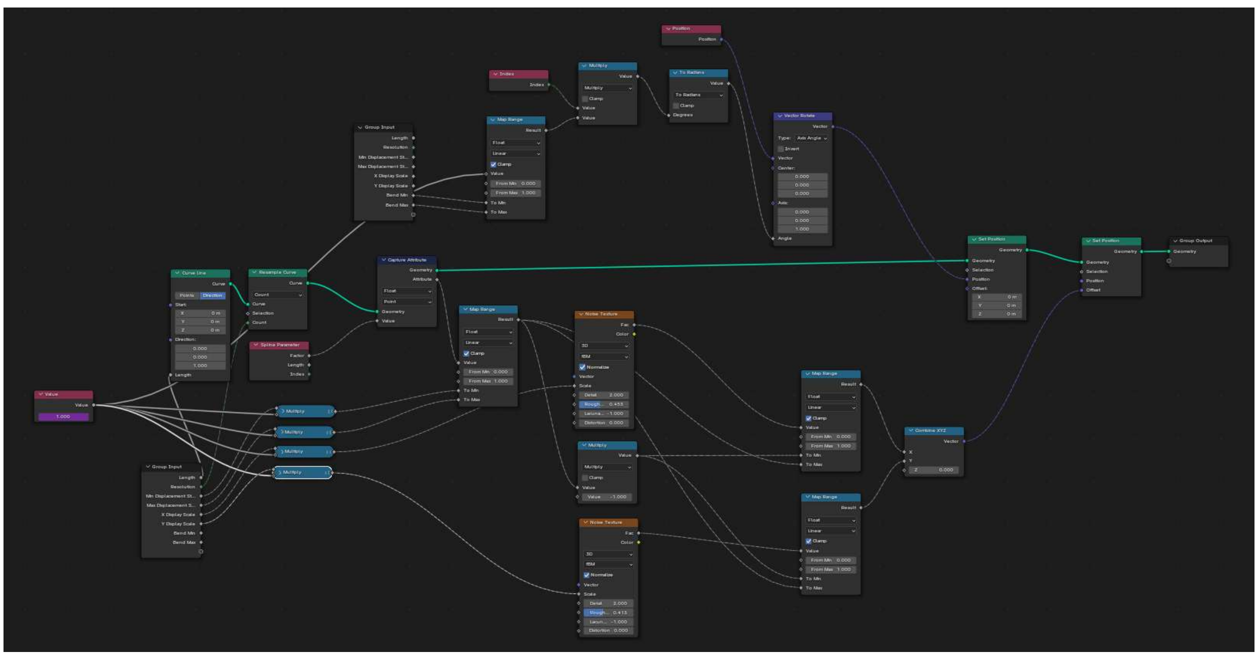



Capture Attribute
Geometry
Attribute
Float
Point
Geometry
Value
Map Range
Result
Float
Linear
Clamp
Value
From Min 0.000
From M... 1.000
To Min
To Max
Noise Texture
Fac
Color
3D
fBM
Normalize
Vector
Scale
Detail 2.000
Roughn... 0.453
Lacun... -1.000
Distorti... 0.000
Multiply
Value
Multiply
Clamp
Value
Value -1.000
Noise Texture
Fac
Color
3D
fBM
Normalize
Vector
Scale
Detail 2.000
Roughn... 0.413
Lacun... -1.000
Distorti... 0.000
Map Range
Result
Float
Linear
Clamp
Value
From Min 0.000
From M... 1.000
To Min
To Max
Combine XYZ
Vector
X
Y
Z 0.000
Map Range
Result
Float
Linear
Clamp
Value
From Min 0.000
From M... 1.000
To Min
To Max

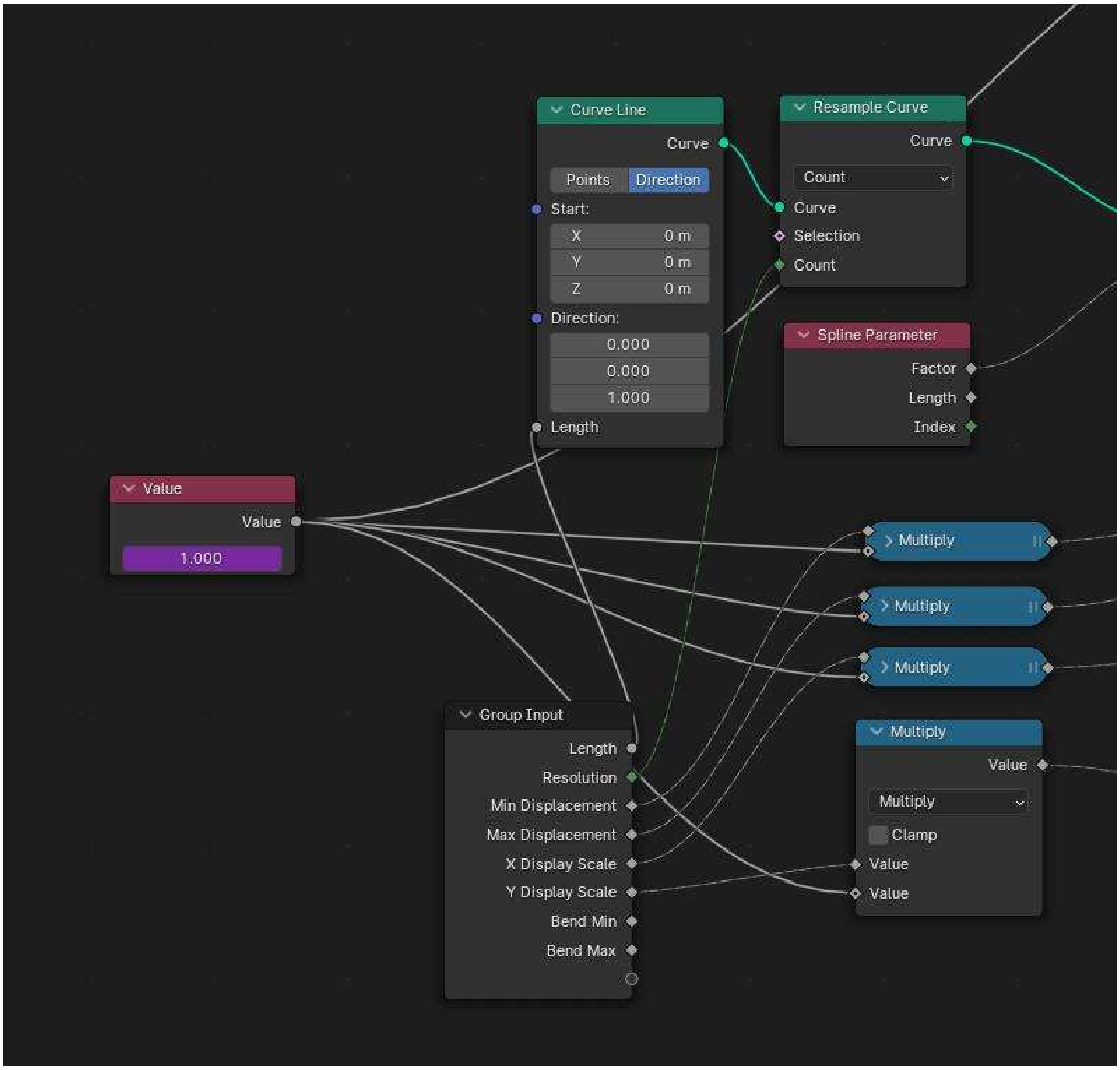

Position
Position
Index
Index
Multiply
Value
Multiply
Clamp
Value
Value
To Radians
Value
To Radians
Clamp
Degrees
Vector Rotate
Vector
Type: Axis Angle
Invert
Vector
Center:
0.000
0.000
0.000
Axis:
0.000
0.000
1.000
Angle
Group Input
Length
Resolution
Min Displacement St...
Max Displacement S...
X Display Scale
Y Display Scale
Bend Min
Bend Max
Map Range
Result
Float
Linear
Clamp
Value
From Min 0.000
From M... 1.000
To Min
To Max
Capture Attribute

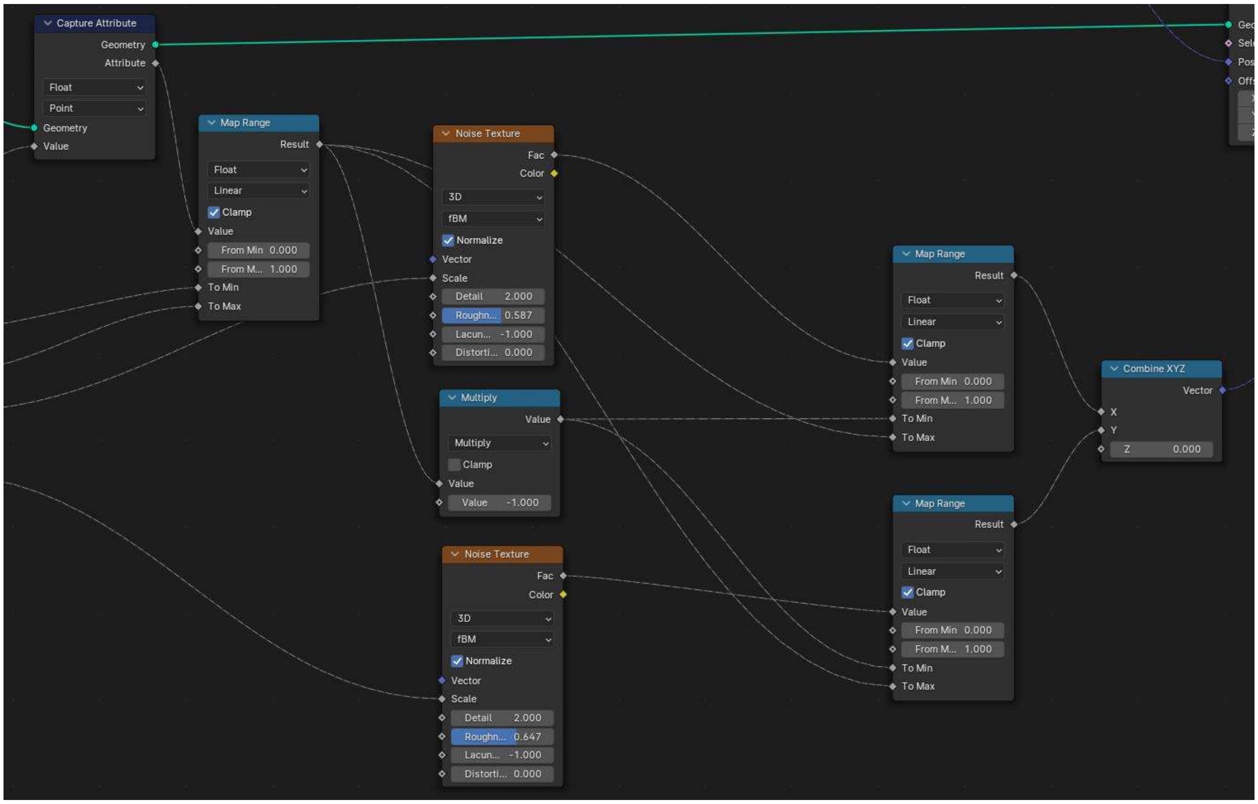


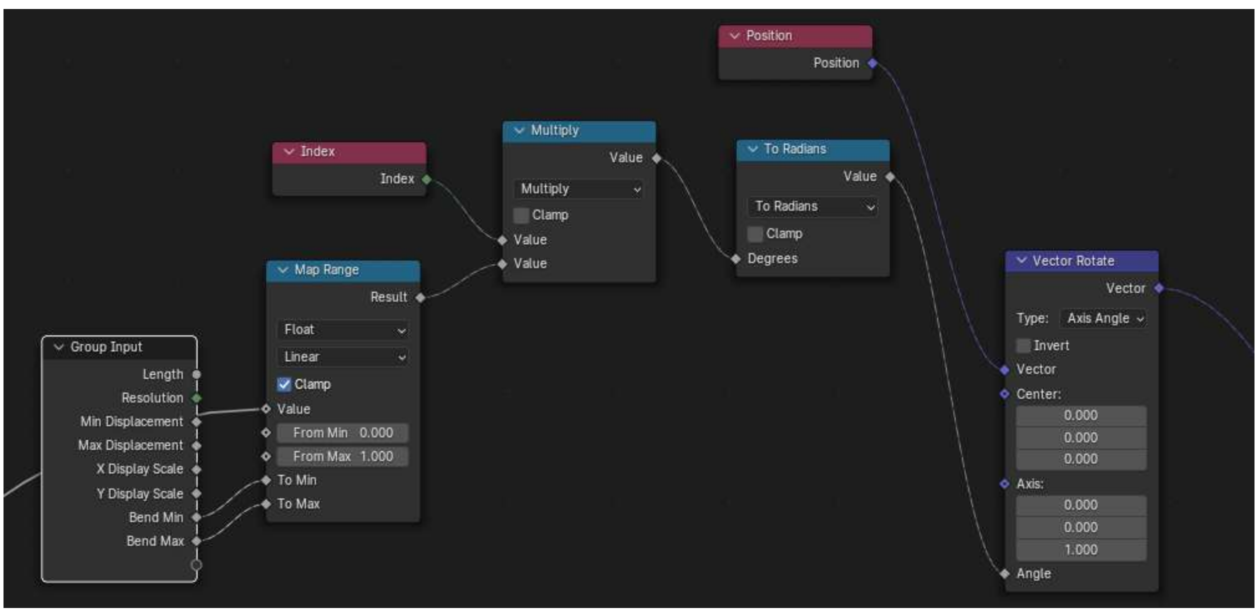
All Together

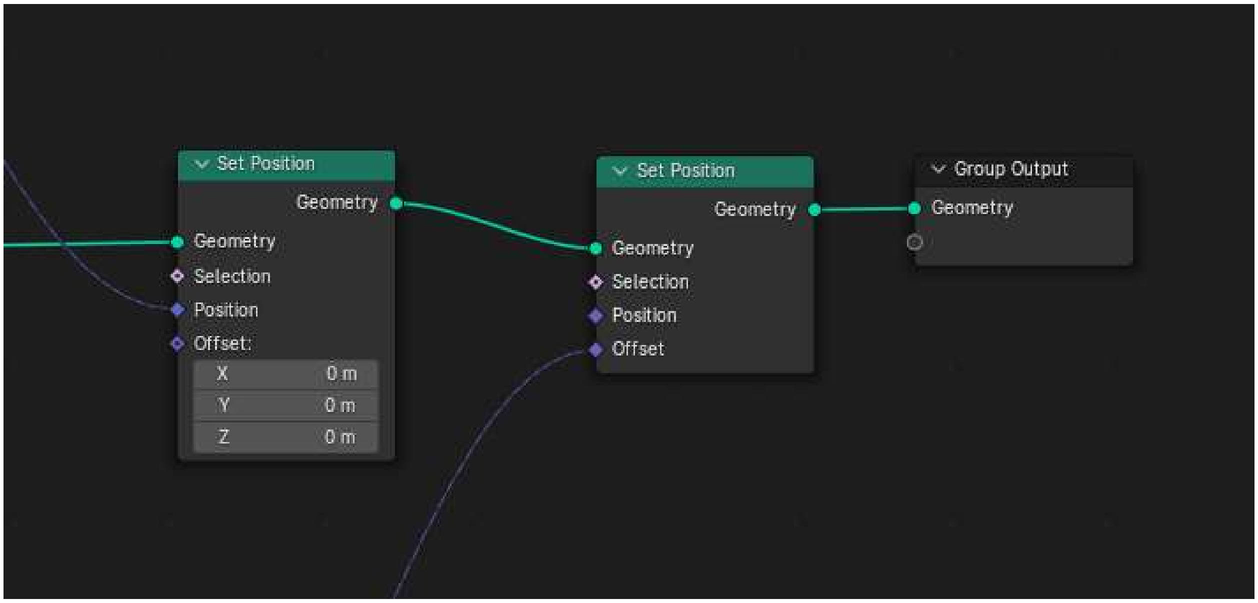


Procedural Twigs

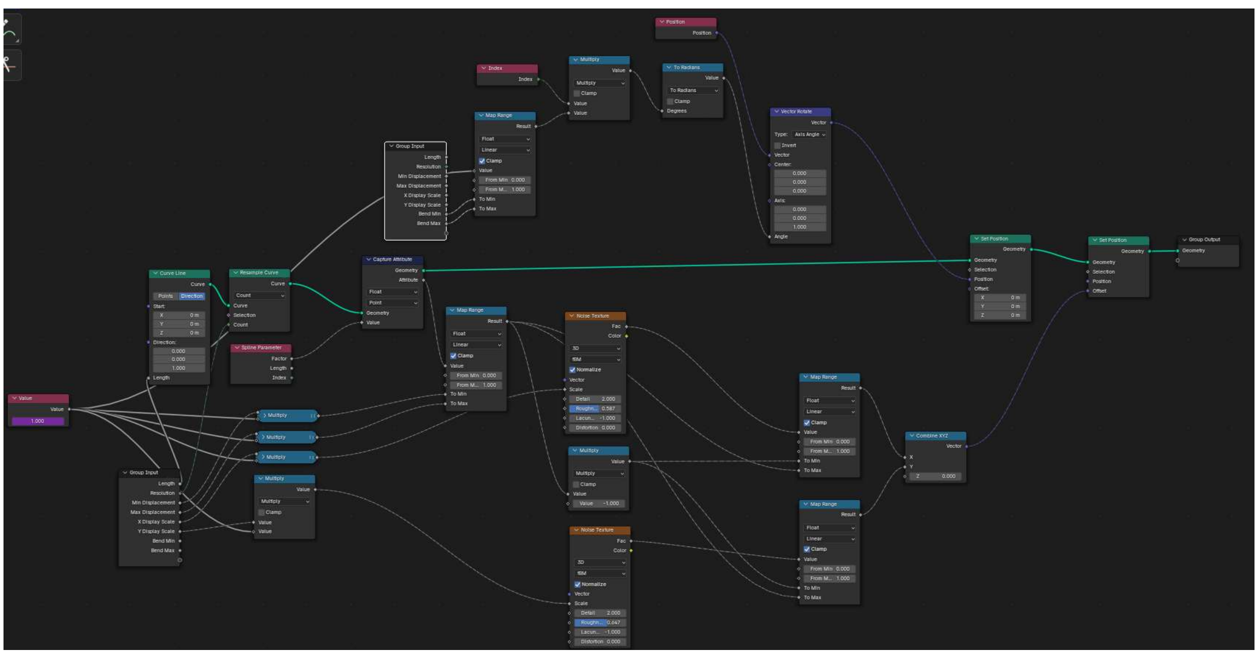



Capture Attribute
Geometry
Attribute
Float
Point
Geometry
Value
Map Range
Result
Float
Linear
Clamp
Value
From Min 0.000
From M... 1.000
To Min
To Max
Noise Texture
Fac
Color
3D
fBM
Normalize
Vector
Scale
Detail 2.000
Roughn... 0.587
Lacun... -1.000
Distorti... 0.000
Multiply
Value
Multiply
Clamp
Value
Value -1.000
Noise Texture
Fac
Color
3D
fBM
Normalize
Vector
Scale
Detail 2.000
Roughn... 0.647
Lacun... -1.000
Distorti... 0.000
Map Range
Result
Float
Linear
Clamp
Value
From Min 0.000
From M... 1.000
To Min
To Max
Map Range
Result
Float
Linear
Clamp
Value
From Min 0.000
From M... 1.000
To Min
To Max
Combine XYZ
Vector
X
Y
Z 0.000


Position
Position
Index
Index
Multiply
Value
Multiply
Clamp
Value
Value
To Radians
Value
To Radians
Clamp
Degrees
Map Range
Result
Float
Linear
Clamp
Value
From Min 0.000
From Max 1.000
To Min
To Max
Group Input
Length
Resolution
Min Displacement
Max Displacement
X Display Scale
Y Display Scale
Bend Min
Bend Max
Vector Rotate
Vector
Type: Axis Angle
Invert
Vector
Center:
0.000
0.000
0.000
Axis:
0.000
0.000
1.000
Angle



All Together



Final Blender Creation

- The roots are just procedural trees that have altered inputs

Final Scene